8 Agreement, case and movement

8.1 Overview

In this chapter, we take a look at the syntax of agreement. We begin by outlining the claim made by Chomsky in recent work that agreement involves a relation between a **probe** and a **goal** (though it should be noted that the term **goal** in this chapter is used in an entirely different way from the term GOAL which was used to denote the thematic role played by a particular kind of argument in relation to its predicate in §7.5). We look at the nature of agreement, and go on to show that nominative and null case-marking involve agreement with T. Finally, we explore the relationship between the [EPP] feature carried by T and agreement, and look at the consequences of this for control infinitives on the one hand and raising infinitives on the other.

8.2 Agreement

In traditional grammars, finite auxiliaries are said to agree with their subjects. Since (within the framework used here) finite auxiliaries occupy the head T position of TP and their subjects are in spec-TP, in earlier work agreement was said to involve a specifier–head relationship (between T and its specifier). However, there are both theoretical and empirical reasons for doubting that agreement involves a spec–head relation. From a theoretical perspective (as we saw in §4.9), Minimalist considerations lead us to the conclusion that we should restrict the range of syntactic relations used in linguistic description, perhaps limiting it to the relation **c-command** created by merger. From a descriptive perspective, a spec–head account of agreement is problematic in that it fails to account for agreement between the auxiliary *are* and the nominal *several prizes* in passive structures such as:

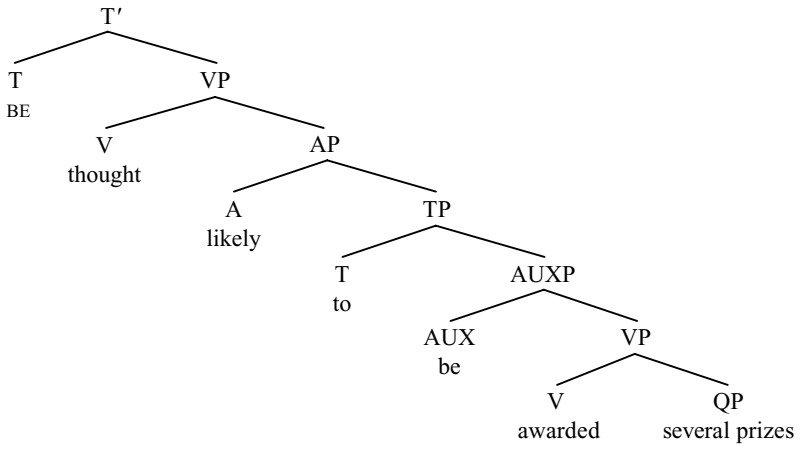
(1) There **are** thought likely to be awarded *several prizes*

Since the auxiliary *are* occupies the head T position of TP in (1) and the expletive pronoun *there* is in spec-TP, a spec–head account of agreement would lead us to expect that *are* should agree with *there*. But instead, *are* agrees with the in situ

complement *several prizes* of the passive participle *awarded*. What is going on here? In order to try and understand this, let's take a closer look at the derivation of (1).

The quantifier *several* merges with the noun *prizes* to form the QP *several prizes*. This is merged as the thematic complement of the passive verb *awarded* to form the VP *awarded several prizes*. This in turn is merged with the passive auxiliary *be* to form the AUXP *be awarded several prizes*. This is then merged with the infinitival tense particle *to*, forming the TP *to be awarded several prizes*. The resulting TP is merged with the raising adjective *likely* to derive the AP *likely to be awarded several prizes*. This AP is subsequently merged with the passive verb *thought* to form the VP *thought likely to be awarded several prizes*. This in turn merges with the passive auxiliary *be*, forming the T-bar shown in simplified form in (2) below (where the notation BE indicates that the morphological form of the relevant item hasn't yet been determined):

(2)



The tense auxiliary [$_T$ BE] needs to agree with an appropriate nominal within the structure containing it. Given Pesetsky's **Earliness Principle** (which requires operations to apply as early as possible in a derivation), T-agreement must apply as early as possible in the derivation, and hence will apply as soon as BE is introduced into the structure. On the assumption that c-command is central to syntactic operations, T will agree with a nominal (i.e. a noun or pronoun expression) which it c-commands. Accordingly, as soon as the structure in (2) is formed, [$_T$ BE] searches for a nominal which it c-commands to agree with.

To use the terminology introduced by Chomsky (1998, 1999, 2001), by virtue of being the highest head in the overall structure at this point in the derivation, BE serves as a **probe** which searches for a c-commanded nominal **goal** to agree with. The only nominal goal c-commanded by [$_T$ BE] within the structure in (2) is the QP *several prizes*: [$_T$ BE] therefore agrees in person and number with *several prizes*, and so is ultimately spelled out as the third-person-plural form *are* in the PF component. Chomsky refers to person and number features together as

φ-features (where φ is the Greek letter *phi*, pronounced in the same way as *fie* in English): using this terminology, we can say that the probe [$_{T}$ BE] agrees in φ-features with the goal *several prizes*. Subsequently, expletive *there* is merged in spec-TP to satisfy the [EPP] requirement for T to project a specifier, and the resulting TP is in turn merged with a null declarative complementiser to form the CP shown in simplified form below (which is the structure of (1) above):

(3) [$_{CP}$ [$_{C}$ ø] [$_{TP}$ There [$_{T}$ are] [$_{VP}$ [$_{V}$ thought] [$_{AP}$ [$_{A}$ likely] [$_{TP}$ to be awarded several prizes]]]]]

However, there are a number of details which we have omitted in (3); one relates to the **case** assigned to the complement (*several prizes*) of the passive participle *awarded*. Although case is not overtly marked on the relevant noun expressions in English, evidence from languages like Icelandic with a richer case system suggests that the complement of a passive participle in finite expletive clauses is assigned nominative case via agreement with T – as the following contrast (from Sigurðsson 1996, p. 12) illustrates:

(4) (a) þad voru lesnar *fjórar bækur*
There were read four$_{NOM.PL}$ books$_{NOM.PL}$

(b) þad var skilað *fjórum bókum*
There was returned four$_{DAT.PL}$ books$_{DAT.PL}$

In (4a), the auxiliary *voru* is a third-person-plural form which agrees with the NOM.PL/nominative plural complement *fjórar bækur* 'four books'. In (4b), the auxiliary is the agreementless form *var* 'was', and the complement of the passive participle is DAT.PL/dative plural. (*Var* is a third-person-singular form, but can be treated as an agreementless form if we characterise agreement by saying that 'An auxiliary is first/second person if it agrees with a first/second-person subject, but third person otherwise; it is plural if it agrees with a plural subject, but singular otherwise.' This means that a third-person-singular auxiliary can arise either by agreement with a third-person-singular expression or – as here – can be a **default** form used as a fall-back when the auxiliary doesn't agree with anything.) Sigurðsson argues that it is an inherent lexical property of the participle *skila* 'returned' that (like around a quarter of transitive verbs in Icelandic) it assigns so-called **inherent** dative case to its complement (see Svenonius 2002a,b on dative complements), and (because it can't agree with a non-nominative complement) the auxiliary surfaces in the agreementless form *var*; by contrast, the participle *lesnar* 'read' in (4a) does not assign inherent case to its complement, and instead the complement is assigned (so-called) **structural** nominative case via agreement with the past-tense auxiliary *voru* 'were'.

Icelandic data like (4) suggest that there is a systematic relationship between nominative case assignment and T-agreement: they are two different reflexes of an **agreement** relationship between a finite T probe and a nominal goal. In consequence of the agreement relationship between the two, the T probe agrees with a nominal goal which it c-commands, and the nominal goal is assigned nominative

case. Accordingly, *several prizes* in (3) receives nominative case via agreement with [$_T$ *are*]. (It should be noted in passing that throughout this chapter, we focus on characterising syntactic agreement. On so-called 'semantic agreement' in British English structures like *The government are ruining the country*, see den Dikken 2001 and Sauerland and Elbourne 2002.)

The approach to case assignment outlined here (in which subjects are assigned nominative case via agreement with a finite T) is different from that outlined in §4.10, where we suggested that subjects are case-marked by a c-commanding C constituent. But in one sense, our revised hypothesis that finite subjects are case-marked by T is consistent with our earlier analysis. In chapter 4, we argued that (in consequence of the Earliness Principle) a noun or pronoun expression is case-marked by the closest case-assigner which c-commands it: since we also assumed in chapter 4 that subjects originate in spec-TP, it was natural to assume that they are case-marked by the closest functional head above them, namely C. But once we move to an analysis like that in chapter 7 in which subjects originate internally within VP, our assumption that they are case-marked by the closest case-assigning head above them opens up the possibility that nominative subjects may be case-marked by T rather than by C – and indeed this is the assumption which we will make from now on (an assumption widely made in current research).

8.3 Feature valuation

Let's think through rather more carefully what it means to say that case is systematically related to agreement, and what the mechanism is by which case and agreement operate. To illustrate our discussion, consider the derivation of a simple passive such as that produced by speaker B below:

(5) SPEAKER A: What happened to the protestors?
SPEAKER B: *They were arrested*

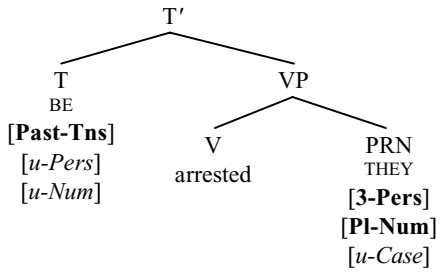
Here, discourse factors determine that a third-person-plural pronoun is required in order to refer back to the third-person-plural expression *the protestors*, and that a past-tense auxiliary is required because the event described took place in the past. So (as it were) the person/number features of *they* and the past-tense feature of *were* are determined in advance, before the items enter the derivation. By contrast the case feature assigned to *they* and the person/number features assigned to *were* are determined via an agreement operation in the course of the derivation: e.g. if the subject had been the singular pronoun *one*, the auxiliary would have been third person singular via agreement with *one* (as in *One was arrested*); and if THEY had been used as the object of a transitive verb (as in *The police arrested them*), it would have surfaced in the accusative form *them* rather than the nominative form *they*.

Generalising at this point, let's suppose that noun and pronoun expressions like THEY enter the syntax with their (person and number) φ-features already

valued, but their case feature as yet **unvalued**. (The notation THEY is used here to provide a case-independent characterisation of the word which is variously spelled out as *they/them/their* depending on the case assigned to it in the syntax.) Using a transparent feature notation, let's say that THEY enters the derivation carrying the features [3-Pers, Pl-Num, u-Case], where *Pers* = person, *Pl* = plural, *Num* = number, and *u* = unvalued. Similarly, let's suppose that finite T constituents (like the tense auxiliary BE) enter the derivation with their tense feature already valued, but their person and number φ-features as yet unvalued (because they are going to be valued via agreement with a nominal goal). This means that BE enters the derivation carrying the features [Past-Tns, u-Pers, u-Num]. In the light of these assumptions, let's see how the derivation of (5B) proceeds.

The pronoun THEY is the thematic complement of the passive verb *arrested* and so merges with it to form the VP *arrested THEY*. This is in turn merged with the tense auxiliary BE, forming the structure (6) below (where already-valued features are shown in **bold**, and unvalued features in *italics*):

(6)

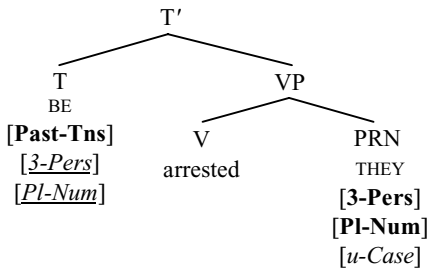


Given Pesetsky's **Earliness Principle**, T-agreement will apply at this point. Let's suppose that agreement in such structures involves a c-command relation between a probe and a goal in which unvalued φ-features on the probe are valued by the goal, and an unvalued case feature on the goal is valued by the probe. (In Chomsky's use of these terms, it is the unvalued person/number features which serve as the probe rather than the item BE itself, but this is a distinction which we shall overlook throughout, in order to simplify exposition.) Since [$_T$ BE] is the highest head in the structure (6), it serves as a probe which searches for a c-commanded goal with an unvalued case feature, and locates the pronoun THEY. Accordingly, an agreement relation is established between the probe BE and the goal THEY. One reflex of this agreement relation is that the unvalued person and number features carried by the probe BE are valued by the goal THEY. Valuation here involves a **Feature-Copying** operation which we can sketch in general terms as follows (where α and ß are two different constituents contained within the same structure, and where one is a probe and the other a goal):

(7) **Feature-Copying**
If α is valued for some feature [F] and β is unvalued for [F] and if β agrees with α, the feature-value for [F] on α is copied onto ß

In consequence of the Feature-Copying operation (7), the values of the person/number features of THEY are copied onto BE, so that the unvalued person and number features [*u-Pers*, *u-Num*] on BE in (6) are assigned the [*3-Pers*, *Pl-Num*] values carried by THEY – as shown in (8) below, where the underlined features are those which have been valued via the Feature-Copying operation (7):

(8)

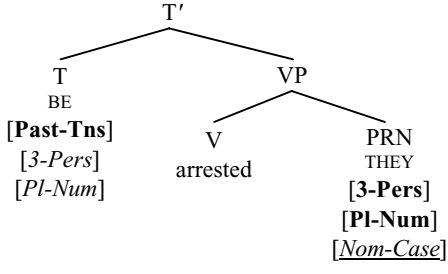


A second reflex of the agreement relation between BE and THEY is that the unvalued case feature [*u-Case*] carried by the goal THEY is valued by the probe BE. Since only auxiliaries with finite (present/past) tense have nominative subjects (and not e.g. infinitival auxiliaries), we can suppose that it is the finite tense features of the probe which are responsible for assigning nominative case to the goal. Accordingly, we can posit that nominative case assignment involves the kind of operation sketched informally below:

(9) **Nominative Case Assignment**
An unvalued case feature on a goal is valued as nominative by a probe carrying finite tense if probe and goal match in φ-features (i.e. in person and number)

Since the person/number features of the probe BE match those of the goal THEY in (8), and since BE carries finite tense (by virtue of its [Past-Tns] feature), the unvalued case feature on THEY is valued as nominative, resulting in the structure shown in (10) below (where the underlined feature is the one valued as nominative in accordance with (9) above):

(10)



Since all the features carried by BE are now valued, BE can ultimately be spelled out in the phonology as the third-person-plural past-tense form *were*. Likewise, since all the features carried by THEY are also valued, THEY can ultimately be spelled out as the third-person-plural nominative form *they*. However, the derivation in (8) is not yet terminated: the [EPP] feature of T will subsequently trigger A-movement

of *they* to become the structural subject of *were*, and the resulting TP *they were arrested ~~they~~* will then be merged with a null declarative complementiser to form the structure *ø they were arrested ~~they~~*: but since our immediate concern is with case and agreement, we skip over these details here.

Although we have given an essentially Chomskyan account of nominative case-marking in (9) and will continue to use it throughout the rest of the book, a theoretically more elegant account would be to make use of Pesetsky and Torrego's assumption (discussed in §6.6) that nominative case is a manifestation of a *tense* feature on T. On this alternative view, the [*u-Case*] feature on THEY in (8) would be replaced by a [*u-Tense*] feature which is valued as [*Past-Tense*] by the Feature-Copying operation in (7), with any (present- or past-) tensed form of the pronoun being spelled out as *they*. This solution is more elegant in two respects. Firstly, it eliminates the need for a **Nominative Case Assignment** operation, since nominative case assignment becomes a tense-copying operation which is simply a particular instance of the **Feature-Copying** operation in (7). Secondly, it avoids a potential violation of a UG principle which Chomsky terms the **Inclusiveness Condition** and which he says (1999, p. 2) 'bars introduction of new elements (features) in the course of a derivation'. Under the analysis sketched in (8), THEY enters the derivation with an *unvalued* case feature which is then assigned the value *nominative* via agreement with a T constituent which has *person, tense and number* features. So it would seem that the value *nominative* is introduced into the derivation via a case-valuation operation like (9), leading to a potential violation of the **Inclusiveness Condition**. By contrast, under the alternative tense-copying analysis of nominative case, no new feature value is introduced into the derivation: instead, the existing [Past] value for the [Tns] feature on T is copied onto the subject.

8.4 Uninterpretable features and feature-deletion

Our discussion of how case and agreement work in a sentence such as (5B) has wider implications. One of these is that items may enter the derivation with some of their features already **valued** and others as yet **unvalued**: e.g. BE enters the derivation in (6) with its tense feature valued, but its φ-features unvalued; and THEY enters with its φ-features valued but its case feature unvalued. This raises the question of which features are initially valued when they first enter the derivation, which are initially unvalued – and why. Chomsky (1998) argues that the difference between valued and unvalued grammatical features correlates with a further distinction between those grammatical features which are **interpretable** (in the sense that they play a role in semantic interpretation), and those which are **uninterpretable** (and hence play no role in semantic interpretation). For example, it seems clear that the case feature of a pronoun like THEY is uninterpretable, since a subject pronoun surfaces as nominative, accusative or genitive depending on the type of [bracketed] clause it is in, without any effect on meaning – as the examples in (11) below illustrate:

(11) (a) It is said [*they* were arrested]
(b) He expected [*them* to be arrested]
(c) He was shocked at [*their* being arrested]

By contrast, the (person/number) φ-features of pronouns are interpretable, since e.g. a first-person-singular pronoun like *I* clearly differs in meaning from a third-person-plural pronoun like *they*. In the case of finite auxiliaries, it is clear that their tense features are interpretable, since a present-tense form like *is* differs in meaning from a past-tense form like *was*. By contrast, the (person/number) φ-features of auxiliaries are uninterpretable, in that they serve purely to mark agreement with a particular nominal. This suggests a correlation such as (12) below between whether or not features are interpretable and whether or not they are initially valued:

(12) **Feature Value Correlation**
(i) Interpretable features enter the derivation already valued
(ii) Features which enter the derivation unvalued are uninterpretable

The correlation between valuedness and interpretability turns out to be an important one. (It should be noted that Chomsky 1998 offers a rather different formulation of (12ii) to the effect that uninterpretable features enter the derivation unvalued, but his claim seems problematic e.g. for languages in which nouns may enter the derivation with an uninterpretable gender φ-feature with a fixed but arbitrary value: e.g. the noun *Mädchen* 'girl' is inherently neuter in gender in German, though it denotes a feminine entity.)

As we saw in the simplified model of grammar which we presented in §1.3, each structure generated by the syntactic component of the grammar is subsequently sent to the **PF component** of the grammar to be **spelled out** (i.e. assigned a **PF representation** which provides a representation of its **Phonetic Form**). If we assume that unvalued features are **illegible** to (and hence cannot be processed by) the **PF component**, it follows that every unvalued feature in a derivation must be valued in the course of the derivation, or else the derivation will **crash** (i.e. fail) because the PF component is unable to **spell out** unvalued features. In more concrete terms, this amounts to saying that unless the syntax specifies whether we require e.g. a first-person-singular or third-person-plural present-tense form of BE, the derivation will crash because the PF component cannot determine whether to spell out BE as *am* or *are*.

In addition to being sent to the PF component, each structure generated by the syntactic component of the grammar is simultaneously sent to the **semantic component**, where it is converted into an appropriate **semantic representation**. Clearly, interpretable features play an important role in the computation of semantic representations. Equally clearly, however, uninterpretable features play no role whatever in this process: indeed, since they are illegible to the semantic component, we need to devise some way of ensuring that uninterpretable features do not input into the semantic component. How can we do this?

Chomsky's answer is to suppose that uninterpretable features are **deleted** in the course of the syntactic derivation, in the specific sense that they are marked as being invisible in the semantic component while remaining visible in the syntax and in the PF component. To get a clearer idea of what this means in concrete terms, consider the uninterpretable nominative case feature on *they* in (5B) *They were arrested*. Since this case feature is uninterpretable, it has to be deleted in the course of the syntactic derivation, so that the semantic component cannot 'see' it. However, the PF component must still be able to 'see' this case feature, since it needs to know what case has been assigned to the pronoun THEY in order to determine whether the pronoun should be spelled out as *they*, *them* or *their*. This suggests the following convention:

(13) **Feature Visibility Convention**
Any uninterpretable feature deleted in the syntax is invisible to the semantic component, but remains visible in the syntactic component and in the PF component

The next question to ask at this juncture is what kind of syntactic operation is involved in the deletion of uninterpretable features. Let's suppose (following Chomsky) that feature-deletion involves the kind of operation outlined informally below (where α and ß enter into an agreement relation, and one is a probe and the other a goal):

(14) **Feature-Deletion**
α deletes any uninterpretable (person/number/case) feature(s) carried by ß if α is φ-complete and if the value(s) of any φ-feature(s) carried by ß match those of the corresponding φ-feature(s) of α

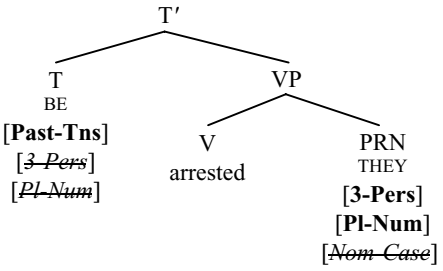
Here, α and ß are two different constituents contained within the same structure, and one is a probe and the other a goal. In a language like English where finite verbs agree with their subjects in person and number (but not gender), ß is **φ-complete** if it carries both person and number features (though in a language like Arabic where finite verbs agree in person, number and gender with their subjects, β is φ-complete if it carries person, number and gender: see Nasu 2001, 2002 for discussion). For ß to delete the person/number/case features of α, the φ-features of ß must **match** the φ-features carried by α. Let's define the relation 'match' in the following terms:

(15) **The relation 'match'**
α and β *match* in respect of some feature [F] either if both have the same value for [F], or if one is valued for [F] and the other unvalued for [F] – but not if they have different values for [F].

To make a rather abstract discussion more concrete, let's consider how feature-deletion applies in the case of our earlier structure (10) above. Here, both BE and THEY are φ-complete, since both are specified for person as well as number. Moreover, the two *match* in respect of their φ-features, since the two have

the same value for person and number (in that both are third person plural). Let's assume that (in consequence of the **Earliness Principle**), feature-deletion applies as early as possible in the derivation, and hence applies at the point where the structure in (10) has been formed. In accordance with Feature-Deletion (14), φ-complete BE can delete the uninterpretable case feature carried by THEY; and conversely φ-complete THEY can delete the uninterpretable person/number features carried by BE. Feature-Deletion therefore results in the structure (16) below (where ~~strikethrough~~ indicates deletion):

(16)



The deleted features will now be invisible in the semantic component – in accordance with (13). The rest of the derivation proceeds as before.

Chomsky sees uninterpretable features as being at the very heart of agreement, and posits (1999, p. 4) that 'Probe and Goal must both be active for Agree to apply' and that a constituent α (whether Probe or Goal) is **active** only if α contains one or more **uninterpretable** features. In other words, it is the presence of uninterpretable features on a constituent that makes it active (and hence able to serve as a probe or goal, and to play a part in feature-valuation and feature-deletion).

As should be obvious, the **Feature-Deletion** operation posited in (14) is very different from the **(Trace) Copy-Deletion** operation we assumed in earlier chapters by which a trace copy of a moved constituent is deleted. Feature-Deletion is an operation which renders the affected features invisible to the semantic component, while leaving them visible to the phonological component. By contrast, Copy-Deletion is an operation which renders traces of moved constituents invisible to the phonological component (in the sense that they are not given any phonetic spellout), while leaving them visible in the semantic component. The reason why traces must remain visible in the semantic component is that they play an important role in semantic interpretation, as we can see in relation to a sentence such as (17a) below, which has the simplified structure (17b) (assuming that *a famous politician* originates as the object of the passive participle *seen* and raises to become the subject of *has*):

(17) (a) A famous politician has not been seen in the Hotel Casanova for weeks
(b) *A famous politician* has not been seen ~~a famous politician~~ in the Hotel Casanova for weeks

(17a) exhibits a **scope ambiguity** in respect of whether *a famous politician* has scope over *not* (so that the sentence is paraphraseable as 'There is a specific

famous politician, Gerry Attrick, who has not been seen in the Hotel Casanova for weeks') or conversely whether *not* has scope over *a famous politician* (so that the sentence is paraphraseable as 'Not a single famous politician has been seen in the Hotel Casanova for weeks'). If the semantic component is able to 'see' traces, it will 'see' the structure represented in skeletal form in (17b) above. One way of handling the scope ambiguity of sentences like (17a) is to posit that scope is defined in terms of c-command and that the scope ambiguity correlates with the fact that in the structure (17b), *not* is c-commanded by (so falls within the scope of) the moved constituent *a famous politician*, but conversely *not* c-commands (and hence has scope over) its trace ~~*a famous politician*~~. A plausible conclusion to draw is that trace-deletion takes place in the phonological component, so that traces remain in the syntax and hence are visible in the semantic component, and can play a role in determining scope in relevant types of structure. The assumption that trace-deletion is a phonological operation is implicit in Chomsky's remark (1999, p. 11) that 'Phonological rules . . . eliminate trace.'

8.5 Expletive *it* subjects

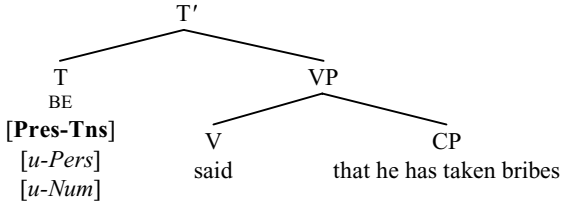
So far, all the constructions we have looked at have involved a finite T agreeing with a noun or pronoun expression which carries interpretable person/number φ-features. However, English has two **expletive pronouns** which (by virtue of being non-referential) carry no interpretable φ-features. One of these is *expletive it* in sentences such as:

(18) (a) *It* is said that he has taken bribes
(b) *It* can be difficult to come to terms with long-term illness
(c) *It*'s a pity that she can't come
(d) *It*'s a long way from here to Lands End

The pronoun *it* in sentences like these appears to be an **expletive**, since it cannot be replaced by a referential pronoun like *this* or *that*, and cannot be questioned by *what*. Let's examine the syntax of expletive *it* by looking at the derivation of a sentence like (18a).

Suppose that we have reached the stage of derivation where the (passive participle) verb *said* has been merged with its CP complement *that he has taken bribes* to form the VP *said that he has taken bribes*. Merging this VP with the tense auxiliary BE forms the structure shown in simplified form below:

(19)



In accordance with Pesetsky's **Earliness Principle**, we might expect T-agreement to apply at this point. Accordingly, the probe BE (which is active by virtue of its uninterpretable person/number φ-features) searches for an active goal to value its unvalued φ-features. It might at first sight seem as if the CP headed by *that* is an appropriate goal, and is a third-person-singular expression which can value the person/number features of BE. However, it seems unlikely that such clauses have person/number features. One reason for thinking this is that even if the *that*-clause in (19) is coordinated with another *that*-clause as in (20) below, the verb BE remains in the singular form *is*:

(20) It is said [*that he has taken bribes* and *that he has embezzled company funds*]

If each of the italicised clauses in (20) were singular in number, we would expect the bracketed coordinate clause to be plural (in the same way as the coordinate structure *John and Mary* is a plural expression in a sentence like *John and Mary are an item*): but the fact that the passive auxiliary *is* remains singular in (20) suggests that the CP has no number properties of its own. Nor indeed does the *that*-clause in (19) have an unvalued case feature which could make it into an active goal, since *that*-clauses appear to be caseless (as argued by Safir 1986), in that a *that*-clause cannot be used in a position like that italicised in (21) below where it would be assigned accusative case by a transitive preposition such as *of*:

(21) *There have been reports of *that he has taken bribes*

If the CP in (19) has no uninterpretable case feature, it is inactive and so cannot value the φ-features of BE.

However, a question we might ask about (19) is whether BE could instead agree with the subject of the *that*-clause, namely *he*: after all, *he* has an uninterpretable case feature (making it active), and is a third-person-singular expression and so could seemingly value the unvalued person/number features of BE. Yet it is clear that BE does not in fact agree with *he*, since if we replace *he* by the first-person-plural subject *we*, BE still surfaces in the third-person-singular form *is* – as (22) below illustrates:

(22) It *is* said [$_{CP}$ that [$_{TP}$ **we** have taken bribes]]

Something, then, must prevent BE from agreeing with *we* – but what? The answer lies in a constraint developed by Chomsky termed the **Phase Impenetrability Condition/PIC**. Since understanding PIC requires a prior understanding of the notion of **phase** developed by Chomsky in recent work (1998, 1999, 2001), let's first take a look at what phases are.

In §1.5 we suggested that a fundamental principle of UG is a **Locality Principle** which requires all grammatical operations to be local. Using the **probe–goal** terminology introduced in this chapter, we can construe this as meaning that all grammatical operations involve a relation between a probe P and a **local** goal G which is sufficiently 'close' to the probe. However, an important question to ask is why probe–goal relations must be local. In this connection, Chomsky (2001,

p. 13) remarks that 'the P, G relation must be local' in order 'to minimise search' (i.e. in order to ensure that a minimal amount of searching will enable a probe to find an appropriate goal). His claim that locality is forced by the need 'to minimise search' suggests a processing explanation: the implication is that the Language Faculty can only process limited amounts of structure at one time – and, more specifically, can only hold a limited amount of structure in its 'active memory' (Chomsky 1999, p. 9). In order to ensure a 'reduction of computational burden' (1999, p. 9) Chomsky proposes that 'the derivation of EXP[ressions] proceeds by *phase*' (ibid.), so that syntactic structures are built up one *phase* at a time. He maintains (2001, p. 14) that 'phases should be as small as possible, to minimise memory'. More specifically, he suggests (1999, p. 9) that phases are 'propositional' in nature, and hence include CPs. His rationale for taking CP to be phases is that CP represents a complete clausal complex (including a specification of **force**).

In what sense do phases ensure that grammatical operations are purely local? The answer given by Chomsky is that any goal within the (c-command) **domain** of the phase (i.e. any goal c-commanded by the head of the phase) is **impenetrable** to further syntactic operations. He refers to this condition as the **Phase Impenetrability Condition/PIC** – and we can state it as follows (cf. Chomsky 2001, p. 5, ex. 6):

(23) **Phase Impenetrability Condition/PIC**
Any goal in the (c-command) domain of a phase head is impenetrable to a probe outside the phase

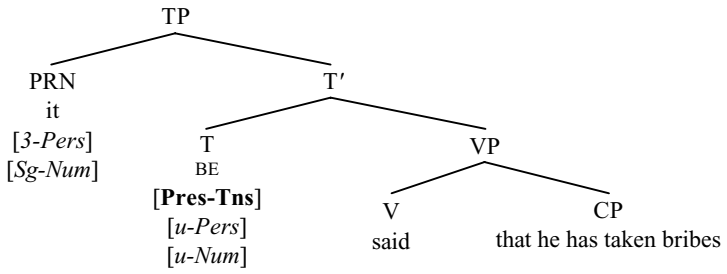
Stated in a form like (23), the relevant condition clearly begs the question of *why* a goal positioned 'below' a phase head should be impenetrable to a probe positioned 'above' the phase. Chomsky's answer (2001, p. 5) is that once a complete phase has been formed, the domain of the phase head (i.e. its complement) undergoes a **transfer** operation by which it is simultaneously sent to the phonological component to be assigned an appropriate phonetic representation, and to the semantic component to be assigned an appropriate semantic representation – and hence no constituent in the relevant domain is thereafter able to undergo any further syntactic operations. So, for example, once the operations which take place on the CP cycle have been completed, the TP which is the domain/complement of the phase head C will be sent to the phonological and semantic components for processing. As a result, TP is no longer accessible in the syntax, and hence neither TP itself nor any constituent of TP can subsequently serve as a goal for a higher probe of any kind in the syntax.

In the light of the Phase Impenetrability Condition (23), let's return to our earlier structure (19) and ask why the auxiliary *is* in the main clause can't agree with the subject *he* of the complement clause. The answer is as follows. The complement clause *that he has taken bribes* is a CP, hence a phase. The domain of that CP (i.e. the constituent which is the complement of the head C of CP) is the TP *he has taken bribes*. This means that neither this TP nor any of its constituents can serve as a goal for a probe outside CP. Since *is* in (19) lies outside the bracketed

CP phase, and *he* lies inside its bracketed TP domain, PIC prevents agreement between the two. (See Polinsky and Potsdam 2001, and Branigan and MacKenzie 2002 for an analysis of apparent long-distance agreement in terms of PIC.)

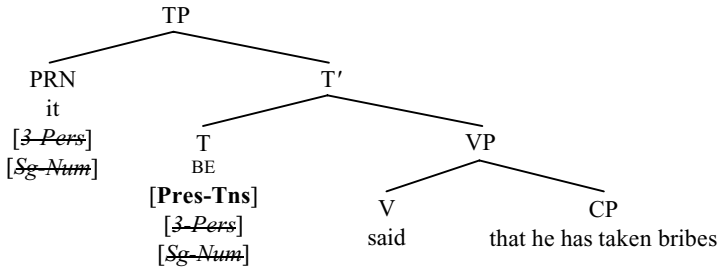
So far, what we have established in relation to the structure in (19) is that BE cannot agree with the *that*-clause because the latter is inactive and has no φ-features or case feature; nor can BE agree with *he*, because PIC makes *he* impenetrable to BE. It is precisely because BE cannot agree with CP or with any of its constituents that expletive *it* has to be used, in order to satisfy the [EPP] requirement of T, and to value the φ-features of T. In keeping with the Minimalist spirit of positing only the minimal apparatus which is conceptually necessary, let's further suppose that expletive *it* has 'a full complement of φ-features' (Chomsky 1998, p. 44) but that (as Martin Atkinson (pc) suggests) these are the only features carried by *it* in its expletive use. More specifically, let's assume that expletive *it* carries the features [third-person, singular-number]. Since expletive *it* is a 'meaningless' expletive pronoun, these features will be *uninterpretable*. Given this assumption, merging *it* as the specifier of the T-bar in (19) above will derive the structure (24) below (with interpretable features shown in bold, and uninterpretable features in italics):

(24)



At this stage in the derivation, the pronoun *it* can serve as a probe because it is the highest head in the structure, and because *it* is active by virtue of its uninterpretable φ-features. Likewise, the auxiliary BE can serve as a goal for *it* because BE is c-commanded by *it* and BE is active by virtue of its uninterpretable φ-features. Feature-Copying (7) can therefore apply to value the unvalued φ-features on BE as third person singular (via agreement with *it*), and Feature-Deletion (14) can apply to delete the uninterpretable φ-features of both *it* and BE, so deriving:

(25)

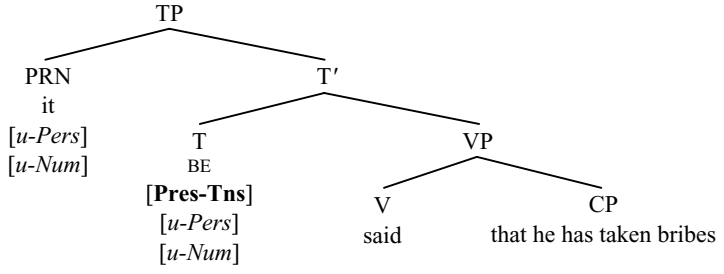


As required, all unvalued features have been valued at this point (BE ultimately being spelled out in the PF component as *is*), and all uninterpretable features deleted. The resulting structure (25) is subsequently merged with a null declarative complementiser. The deleted uninterpretable person/number features of *it* and BE will be visible in the PF component and the syntax, but not in the semantic component; the undeleted [**Pres-Tns**] feature of BE will be visible in all three components. Hence, BE will be spelled out as *is* in the PF component, since the phonology can 'see' the third-person, singular-number, present-tense features carried by BE.

There are two particular features of the analysis outlined above which merit further comment. One is that we have assumed that expletive *it* carries person and number features, but no gender feature and no case feature. While *it* clearly carries an interpretable (neuter/inanimate) gender feature when used as a referential pronoun (e.g. in a sentence like *This book has interesting exercises in it*, where *it* refers back to *this book*), it has no semantic interpretation in its use as an expletive pronoun, and so can be assumed to carry no interpretable gender feature in such a use. The reason for positing that expletive *it* is a caseless pronoun is that it is already active by virtue of its uninterpretable φ-features, and hence does not 'need' a case feature to make it active for agreement (unlike subjects with interpretable φ-features). Some suggestive evidence that expletive *it* may be a caseless pronoun comes from the fact that it has no genitive form *its* – at least for speakers like me who don't say (e.g.) **He was annoyed at its raining*.

A further assumption worth commenting on is that we have assumed that expletive *it* is intrinsically third person singular, and that this is why BE ends up in the third-person-singular form *is* in sentences like (18a) *It is said that he has taken bribes*. However, if we were to accept Chomsky's view that all uninterpretable features enter the derivation unvalued, we'd have to say that the pronoun *it* enters the derivation with unvalued person/number features, as in (26) below:

(26)



The obvious question which such an analysis would raise is how a pronoun with unvalued person/number features can value the unvalued person/number features of T (and conversely). To answer this question, we'd have to invoke default number/person valuation conditions such as those italicised below:

(27) **Person/Number Valuation**
Where α and β enter into an agreement relation and β has one or more unvalued φ-features which match corresponding φ-features carried by α, α values β as:
(i) first person if α is first person, second person if α is second person, and *third person otherwise*
(ii) plural if α is plural and *singular otherwise*

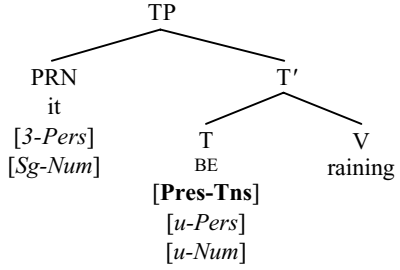
(27i) would ensure that the person features of both *it* and BE are assigned the default (*otherwise*) value [*3-Pers*]; and (27ii) would ensure that the number features of both *it* and BE are assigned the default value [*Sg-Num*]. We will not attempt to choose between the analyses in (24) and (26) here, but for concreteness we will henceforth assume (24).

Let's now turn to consider the question of how we handle sentences like the following, which contain so-called *weather it*:

(28) (a) It is raining (b) It has been snowing

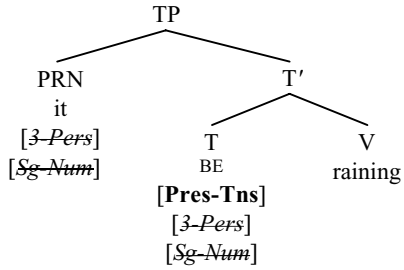
One way of analysing a sentence like (28a) is to treat RAIN as a predicate which has no θ-marked argument, and to take *it* to be a non-referential (expletive) pronoun. This would mean that the first stage in the derivation of (28a) is for the tense auxiliary BE to be merged with the verb RAIN (which is ultimately spelled out as the form *raining* because the progressive auxiliary BE requires a complement headed by a verb in the *ing*-form). Merging expletive *it* as the specifier of the resulting T-bar *BE raining* would derive the structure (29) below, if we assume (for expository purposes) that *it* is intrinsically third person singular:

(29)



At this point, *it* is the highest head in the overall structure, and is active (by virtue of its uninterpretable φ-features) and so can serve as a probe. [$_T$ BE] is c-commanded by *it* and is also active (by virtue of its own uninterpretable φ-features), and so can serve as a goal for the probe *it*. Accordingly, the unvalued person/number features on BE are valued via the **Feature-Copying** operation (7), with the result that the φ-features of BE are assigned the same values as those of *it*. Since both *it* and BE are φ-complete (by virtue of carrying both person and number features), and since their φ-features have matching values, each can delete the uninterpretable φ-features of the other in accordance with **Feature-Deletion** (14), so deriving:

(30)



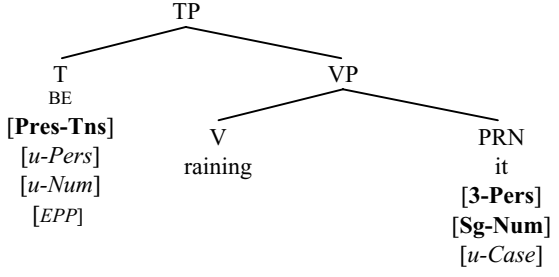
The deleted uninterpretable person/number features of *it* and BE will be visible in the PF component and the syntax, but not in the semantic component; the undeleted [**Pres-Tns**] feature of BE will be visible in all three components. Hence, BE will be spelled out as *is* in the PF component, since PF can 'see' the third-person, singular-number, present-tense features carried by BE. The resulting TP will subsequently be merged with a null declarative-force complementiser.

A key assumption in the analysis outlined above is that expletive *it* is a meaningless 'filler', and so non-referential. However, this assumption would seem to be called into question by the observation that expletive *it* can serve as the antecedent of PRO: cf.

(31) (a) It sometimes rains after PRO snowing (Chomsky 1981, p. 324)
(b) It can seem that someone is guilty without [PRO seeming that they actually committed the crime] (Williams 1994, p. 91)

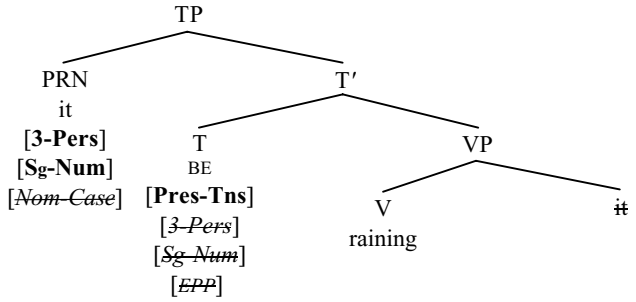
If we suppose that only a referential pronoun can serve as the controller of PRO, a plausible conclusion to draw is that expletive *it* is referential (in a sense made precise by Chomsky 1981, who suggests that expletive *it* is a **quasi-argument**). And if weather *it* in sentences like (28a,b) is referential, it is also plausible to suppose that it is initially merged as a (quasi-)argument of the weather predicate with which it is associated. If we suppose that weather verbs like *rain/snow* are unaccusative (as is suggested by the fact that in Italian they can be used with the auxiliary *essere* 'be' in perfect-participle forms), this would mean that *it* in (28a,b) originates as the complement of the verb *rain/snow*. If weather *it* is indeed referential and if this means that its person/number features are interpretable, then it follows that weather *it* will also need an unvalued case feature to make it active in the syntax. Assuming all of this, we will have the structure shown in simplified form in (32) below at the stage when T is merged with its complement:

(32)



The unvalued person/number features of BE will be valued as third person singular in accordance with **Feature-Copying** (7), and deleted in accordance with Feature-Deletion (14). The unvalued case feature on *it* will be valued as nominative by Nominative Case Assignment (9), and deleted by Feature-Deletion (14). The [EPP] feature of T will simultaneously trigger movement of *it* (which is active by virtue of its unvalued case feature) to spec-TP, so deriving the structure in (33) below (simplified by not showing features carried by the trace copy of *it*):

(33)

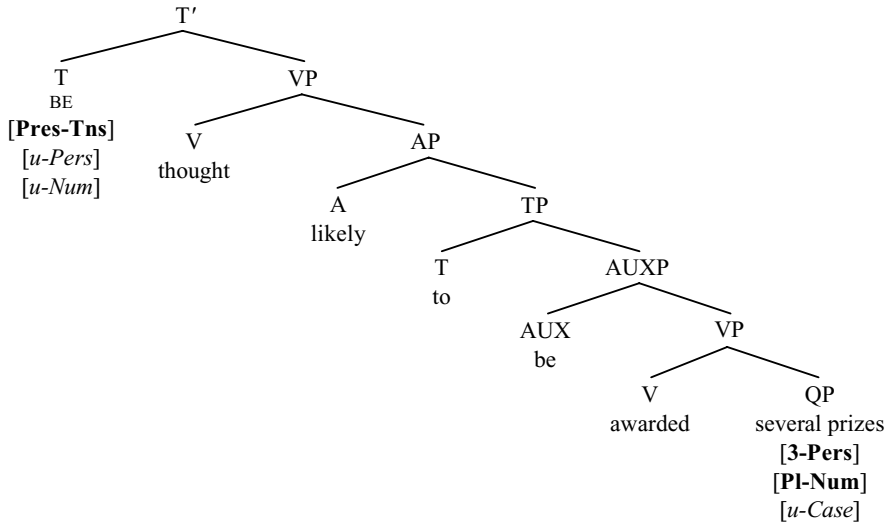


As required, all uninterpretable features have been deleted, so the resulting derivation is convergent. If the analysis of expletive *it* outlined here is along the right lines it suggests that (contrary to what we assumed earlier) expletive *it* is not a pure 'dummy' element inserted in spec-TP to satisfy the [EPP] requirement of T, but rather is a (quasi-)argument which originates internally within VP. Of course, if expletive *it* carries case, we have to ask why (as noted above) it has no genitive form: however, this is arguably just a lexical idiosyncrasy, since even in its referential use *it* has no strong genitive form (Peter Evans (pc) points out), as we see from the ungrammaticality of ?**Its watering the flowers is something I don't like about my cat*.

8.6 Expletive *there* subjects

Having looked at the syntax of expletive *it* in the previous section, we now turn to look at expletive *there*. As a starting point for our discussion, we'll go back to the very first sentence we looked at in this chapter, namely (1) *There are thought likely to be awarded several prizes*. Let's suppose that the derivation proceeds as before, until we reach the stage in (2) above. However, let's additionally assume that *several prizes* carries interpretable φ-features (marking it as a third-person-plural expression) and an uninterpretable (and unvalued) case-feature. Let's also assume (as in earlier discussions) that BE carries an interpretable present-tense feature, and uninterpretable (and unvalued) φ-features. This being so, the structure formed when BE is merged with its VP complement will be that shown in simplified form below:

(34)



Given the **Earliness Principle**, T-agreement will apply at this point in the derivation. Because BE is the highest head in the structure (in that it is the only head in the structure which is not c-commanded by another head), and because BE is active (by virtue of its uninterpretable φ-features), BE serves as a probe which searches for a nominal goal within the structure containing it. The nominal *several prizes* can serve as a goal for the probe BE, since *several prizes* is active by virtue of carrying an uninterpretable case feature. By application of Feature-Copying (7), the unvalued person and number features on BE are given the same values as those on *several prizes* – as shown in simplified form in (35) below:

(35) [BE] thought likely to be awarded [several prizes]
[**Pres-Tns**] [**3-Pers**]
[*3-Pers*] [**Pl-Num**]
[*Pl-Num*] [*u-Case*]

By application of Nominative Case Assignment (9), the unvalued case feature of the goal *several prizes* in (35) is assigned the value nominative as shown in (36) below, since the probe BE carries finite tense (more specifically, present tense), and since the probe [BE] and the goal *several prizes* have matching φ-feature values because both are third person plural:

(36) [BE] thought likely to be awarded [several prizes]
[**Pres-Tns**] [**3-Pers**]
[*3-Pers*] [**Pl-Num**]
[*Pl-Num*] [*Nom-Case*]

Via Feature-Deletion (14), the probe BE deletes the uninterpretable nominative case feature on *several prizes*, since BE is φ-complete (by virtue of carrying both person and number features) and the φ-features of the probe BE match those of the goal *several prizes*. Conversely, via the same Feature-Deletion operation

(14), the goal *several prizes* deletes the uninterpretable person/number features on the probe BE, since the goal is φ-complete (carrying both person and number features), and probe and goal have matching φ-feature values. Feature-Deletion yields:

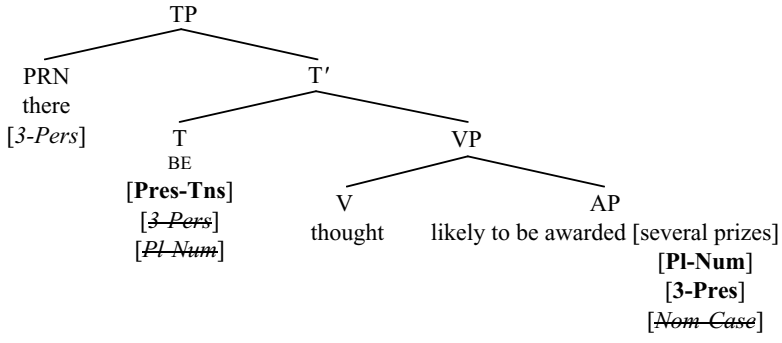
(37)

[BE]	thought likely to be awarded [several prizes]	
[Pres-Tns]		**[3-Pers]**
[~~*3-Pers*~~]		**[Pl-Num]**
[~~*Pl-Num*~~]		[~~*Nom-Case*~~]

We have thus deleted all uninterpretable case/agreement features on both probe and goal, as required.

However, BE also has an [EPP] feature (not shown above) requiring it to project a structural subject. In (1) *There are thought likely to be awarded several prizes*, the [EPP] requirement of [$_T$ BE] is satisfied by merging expletive *there* in spec-TP. Let's assume that (like expletive *it*), expletive *there* carries no case feature (and hence has no genitive form, as we see from the ungrammaticality of **She was upset by there's being nobody to help her*). More precisely, let's follow Chomsky (1998, 1999, 2001) in positing that the only feature carried by expletive *there* is an uninterpretable person feature, and let's further suppose that *there* is intrinsically third person (consistent with the fact that a number of other words beginning with *th-* are third person – e.g. *this*, *that*, *these*, *those* and *the*). Accordingly, merging *there* in spec-TP will derive the structure shown in abbreviated form below:

(38)



The pronoun *there* serves as a probe because it is the highest head in the structure, and because it is active by virtue of carrying an uninterpretable third-person φ-feature. It therefore searches for a c-commanded goal to agree with. Let's suppose that agreement (of the kind we are concerned with here) involves a T–nominal relation (i.e. a relation between T and a noun/pronoun expression): this being so, *there* (being a pronominal probe) will search for an active T constituent to serve as its goal, and find [$_T$ BE]. BE is an active goal for the probe *there* in (38) because *be* contains uninterpretable person/number features: these have been marked as invisible to the semantic component (via Feature-Deletion), but remain visible

and active in the syntax in accordance with the Feature Visibility Convention (13). Accordingly, Feature-Deletion (14) applies, and the goal BE deletes the matching uninterpretable third-person feature carried by the probe *there*. This is possible because *there* is active as a probe and BE is active as a goal (as we have just seen), and because the goal BE is φ-complete (having both person and number features), and the third-person feature carried by the probe *there* matches the third-person feature carried by the goal BE. Deleting the uninterpretable person feature of *there*, and merging the resulting TP with a null complementiser carrying an interpretable declarative force feature [**Dec-Force**], derives the CP shown in skeletal form below:

(39)

ø	there	BE	thought likely to be awarded [several prizes]
[**Dec-Force**]	[~~*3-Pers*~~]	[**Pres-Tns**]	[**3-Pers**]
		[*3-Pers*]	[**Pl-Num**]
		[~~*Pl-Num*~~]	[~~*Nom-Case*~~]

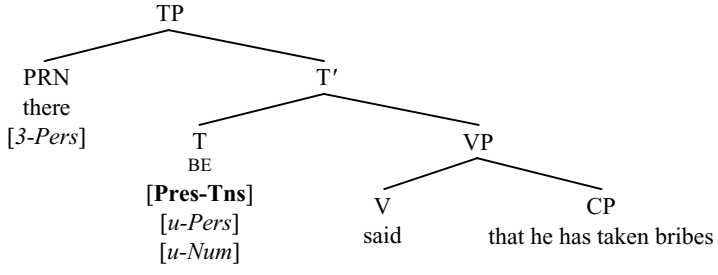
Only the bold-printed interpretable features will be processed by the semantic component, not the barred italicised uninterpretable features (since these have all been deleted and deletion makes features invisible to the semantic component, while leaving them visible to the syntactic and phonological components); both the interpretable and uninterpretable features will be processed by the phonological component where BE will be spelled out as *are*. (On colloquial structures like *There's lots of people in the room*, see den Dikken 2001.)

An important question to ask in the context of our discussion of expletive *it* in the previous section and expletive *there* in this section is what factors determine the choice of expletive in a particular sentence. In this connection, let's ask why expletive *there* can't be used in place of expletive *it* in sentences like (40b) below:

(40) (a) *It* is said that he has taken bribes
(b) **There* is said that he has taken bribes

Let's suppose that merging BE with the VP headed by the verb *said* forms the structure shown in (19) above, and that subsequently merging *there* in spec–TP derives the structure shown in (41) below:

(41)



Because it is the highest head in the structure, and because it is active by virtue of its uninterpretable person feature, *there* serves as a probe. BE serves as the goal for

there because BE is c-commanded by *there*, and BE itself is active by virtue of its uninterpretable person/number features. Via Feature-Copying (7), the unvalued person feature of BE will be assigned the same third-person value as *there* – as shown in schematic form below:

(42) there BE said that he has taken bribes
[*3-Pers*] [**Pres-Tns**]
[*3-Pers*]
[*u-Num*]

Via Feature-Deletion (14), BE can delete the uninterpretable person feature of *there*, because BE is φ-complete and the person features of BE and *there* have matching values. However, *there* cannot delete the person feature of BE, since *there* is φ-incomplete (in that it has person but not number), and only a φ-complete α can delete one or more features of ß. Accordingly, the structure which results after Feature-Deletion applies is:

(43) there BE said that he has taken bribes
[~~*3-Pers*~~] [**Pres-Tns**]
[*3-Pers*]
[*u-Num*]

However, the resulting derivation will ultimately crash, for two reasons. Firstly, the number feature on BE has remained unvalued, and the PF component cannot process unvalued features. And secondly, the uninterpretable person and number features on BE have not been deleted, and the semantic component cannot process uninterpretable features. In other words, our assumptions about the differences between expletive *it* and expletive *there* allow us to provide a principled account of why (40a) *It is said that he has taken bribes* is grammatical, but (40b) **There is said that he has taken bribes* is not.

Now let's ask why expletive *it* can't be used in place of *there* in a sentence like (44b) below:

(44) (a) *There* are thought likely to be awarded several prizes
(b) **It* is thought likely to be awarded several prizes

One way of answering this question is by making the assumption outlined below:

(45) **EPP Generalisation**
When T carries an [EPP] feature, this can be deleted
(i) by merging expletive *there* in spec-TP if T c-commands a matching indefinite goal (i.e. an indefinite noun or pronoun expression which matches T in person/number)
or (ii) by merging expletive *it* in spec-TP if T c-commands no matching goal
or (iii) by moving the closest matching active goal c-commanded by T into spec-TP

The requirement in (45iii) for T to attract the closest matching goal is a consequence of the **Attract Closest Principle**. (45i) stipulates the indefiniteness

requirement without explaining it. An interesting possibility to explore would be that in expletive *there* structures, the associate is indefinite because it has no person properties, so that *there* is inserted in order to value the person properties of T (though see Frampton and Gutmann 1999 for an alternative explanation. See also Lasnik 2001 on the nature of EPP.)

It follows from (45) that in structures like (34) where [$_T$ BE] c-commands (and agrees in person and number with) an indefinite nominal (*several prizes*), expletive *there* can be used but not expletive *it*, so deriving (44a) *There are thought likely to be awarded several prizes*. Conversely in structures like (19) where there is no matching goal accessible to the probe [$_T$BE], *it* can be used but not *there* – so deriving (18a) *It is said that he has taken bribes*. It also follows from (45) that neither expletive can be used in structures like the following:

(46) (a) *There was impeached the president
(b) *It was impeached the president

This is because *was* in (46) c-commands and agrees in person and number with the definite goal *the president*, so that the conditions for the use of either expletive in (45i,ii) are not met. The only way of deleting the [EPP] feature of T in such a case is to passivise the definite DP *the president*, so deriving:

(47) The president was impeached

So, we see that the **EPP Generalisation** in (45) provides a descriptively adequate characterisation of data like (40), (44), (46) and (47). (See Bowers 2002 for an alternative account of the *there/it* distinction in expletives.)

However, our so-called 'generalisation' in (45) is little more than a descriptive stipulation, and begs the question of why the relevant restrictions on the use of expletives should hold. A preferable solution would be to see the choice between expletive *there* and expletive *it* as one rooted in UG principles. Reasoning along these lines, one possibility would be to posit that economy considerations dictate that we use an expletive carrying as few uninterpretable features as possible. In a structure like (19), the expletive has to serve two functions: (i) to satisfy the [EPP] requirement for T to have a specifier with person and/or number properties; and (ii) to value the unvalued person/number features of [$_T$ BE]. Hence only expletive *it* can be used, since this carries both person and number. But in a structure like (2), the expletive is not needed to value the person/number features of [$_T$BE] since these are valued by *several prizes*; rather, the expletive serves only to satisfy the requirement for T to have a specifier with person and/or number features. In this situation, we might suppose, *there* is preferred to *it* because *there* carries only person, and economy considerations dictate that we use as few uninterpretable features as possible.

Throughout this section, we have followed Chomsky in assuming that expletive *there* is a 'dummy' pronoun directly merged in spec-TP. However, just as there are some who believe that expletive *it* originates internally within VP, so too there are some who believe that expletive *there* originates internally within VP,

perhaps with a locative function (as suggested by Moro 1997). Bowers (2002, p. 195) argues that the ungrammaticality of transitive expletive structures such as the following in Standard English:

(48) *There has [someone eaten a bagel]

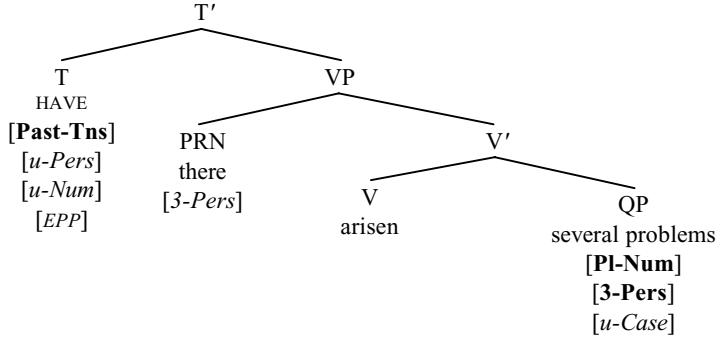
cannot be accounted for under Chomsky's spec-TP analysis of expletive *there*, since nothing in the spec-TP account prevents *has* from agreeing with (and assigning nominative case to) *someone*, with expletive *there* being inserted in spec-TP in order to delete the [EPP] feature of T. A principled way of ruling out sentences like (48), Bowers argues, is by supposing that expletive *there* originates in spec-VP as a nonthematic subject – and hence it can only occur in intransitive VPs which (by their very nature) do not have a thematic subject.

We can illustrate how such an analysis might work in terms of a sentence such as (49) below:

(49) There have arisen several problems

The verb *arise* is an unaccusative predicate which projects a complement, but no thematic subject. Precisely because it projects no thematic subject, it can project an expletive subject (on Bowers's assumption that a predicate only allows an expletive subject if it has no thematic subject). This means that expletive *there* will initially be merged as the specifier of the VP headed by *arisen* in (49), so that at the stage of derivation when HAVE is merged with its VP complement, we have the structure shown in (50) below:

(50)



[$_T$ HAVE] will then probe for active matching goals which carry person and/or number features, and locates *two* such active goals, *there* and *several problems*. Accordingly, T simultaneously agrees with both *there* and *several problems* – resulting in **multiple agreement** (i.e. agreement between a probe and more than one matching goal). Clearly, this will only yield a successful outcome if the associate carries the same third-person feature as *there* – thereby accounting for the observation by Sigurðsson (1996) that expletive associates must be third-person expressions. The [EPP] feature of T simultaneously attracts the closest active goal, so triggering movement of *there* to spec-TP. We can assume that all the

various operations affecting a given probe (like T in (50) above) apply simultaneously, so that agreement with *there*, agreement with *several problems* and movement of *there* to spec-TP all apply at the same time.

A potential problem posed by analysing expletive *there* as a (perhaps locative) quasi-argument initially merged in spec-VP is that (unlike expletive *it* in sentences such as (31) above) expletive *there* cannot serve as a controller for PRO – as we see from (51) below:

(51) There occurred three more accidents without [there/*PRO being any medical help available on the premises] (Haegeman 1994, p. 279)

If quasi-arguments have the property of being able to serve as controllers of PRO, sentences like (51) might be thought to argue against a VP-internal origin for expletive *there*. However, a straightforward way of accounting for the contrast between (31) and (51) is to suppose that PRO requires an antecedent with both person and number features, and that expletive *it* carries both of these features, but expletive *there* carries only person.

A VP-internal analysis of expletives like that in (50) also offers significant theoretical advantages over Chomsky's TP analysis, in that it provides us with a way of avoiding two potentially problematic aspects of Chomsky's analysis. One is that although a probe is generally the *head* of a containing projection (so that the head T of TP is the probe in most of the structures we have looked at in this chapter), Chomsky's TP analysis of expletives has to stipulate that a *specifier* can also serve as a probe when it is an expletive pronoun like *there/it* – a claim which is hard to square with his (2001) view that specifier–head agreement should be eliminated from the set of operations permitted by UG. By contrast, the VP-internal analysis of expletives allows us to maintain the stronger claim that only a *head* can be a probe. A second feature of Chomsky's analysis (illustrated in (38) above) is that he needs to assume that a T which has already had its person/number features valued and deleted by agreement with an indefinite associate can nonetheless serve as an active goal for agreement with an expletive probe. One way round this problem (suggested by Pesetsky and Torrego 2001) is to say that the relevant features are **marked for deletion** (or metaphorically speaking, sentenced to death) on the TP cycle, but not actually deleted (or, metaphorically speaking, executed) until a later stage of derivation (at the end of the CP cycle/phase). However, any such intrinsically undesirable splitting of deletion into two processes can be avoided under the account suggested here, which allows us to posit a unitary treatment of deletion along the lines of (52) below:

(52) **Feature Inactivation Hypothesis**
An uninterpretable feature becomes inactive in the syntax (and invisible to the semantic component) immediately it is deleted

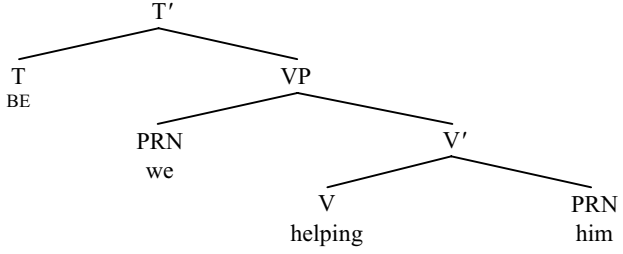
However, it should be noted that while (52) is compatible with a VP analysis of expletives, it is not compatible with Chomsky's claim that expletive *there* is directly merged in spec-TP.

A crucial premise of our alternative account of expletives is that in structures like (50), a T-probe can agree simultaneously with multiple goals (so that *have* simultaneously agrees in person with *there* and in person and number with *several problems*). However, this assumption raises interesting questions about how agreement works in transitive sentences like:

(53) We are helping him

Given the assumptions we are making here, (53) will be derived as follows. The verb *helping* merges with its THEME complement *him* to form the V-bar *helping him*. This V-bar is in turn merged with its AGENT argument *we* to form the VP *we helping him*. The resulting VP is then merged with a present-tense T constituent to form the T-bar shown in simplified form in (54) below:

(54)



Given the Earliness Principle, T will serve as a probe at this point and look for one or more nominal goals to value (and delete) its unvalued person/number features. However, if (as we assumed in our discussion of (50) above) a probe can agree with multiple goals, an important question to ask is why T can't agree with both the subject *we* and the complement *him*. If (contrary to fact) multiple agreement were permitted in structures like (54), it would cause the derivation to crash because the person/number features of BE would have to be valued as first person plural in order to agree with the subject *we* and as third person singular in order to agree with the object *him*, and this would clearly lead to conflicting requirements on how the person/number features of BE should be valued. In reality, T agrees with the subject *we* in transitive structures like (54) and not with the object *him*. But in a framework which allows a probe to agree with multiple goals, how can we rule out agreement between T and the object of a transitive verb?

One answer to this question is provided by the Phase Impenetrability Condition, which we formulated in (23) above in the manner set out in (55) below:

(55) **Phase Impenetrability Condition/PIC**
Any goal c-commanded by a phase head is impenetrable to any probe outside the phase

In our earlier discussion of PIC in §8.5, we noted Chomsky's (1999, p. 9) claim that phases are 'propositional' in nature, and that accordingly CPs are phases. However, Chomsky claims that **transitive verb phrases** (but not intransitive VPs) are also propositional in nature and hence phases, by virtue of the fact

that transitive VPs contain a complete thematic (argument structure) complex, including an external argument in spec-VP.

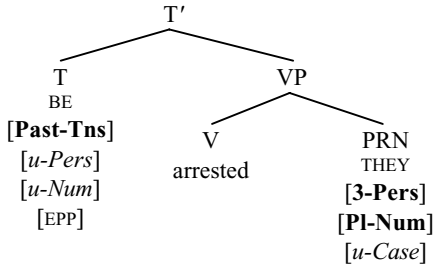
If transitive VPs are phases, and PIC allows only constituents on the **edge** (i.e. in the head or specifier position) of a phase to be accessible to a higher probe, it follows that in a structure like (54) above, the T constituent BE will only be able to agree with the subject *we* on the edge of the transitive VP phase, not with the object *him* which lies within the (c-command) domain of the transitive phase head *helping*. By contrast, in expletive structures like (50), PIC will not prevent the T constituent *have* from agreeing with both *there* and *several problems*, since the VP headed by the unaccusative verb *arisen* is intransitive (its specifier *there* not being an external argument but rather being a non-referential expletive pronoun).

If we adopt the Feature Inactivation Hypothesis (52), there is also another way in which we can prevent agreement between T and the object of a transitive verb in structures like (54). If the object *him* enters the derivation with an unvalued case feature and (in accordance with the Earliness Principle) the relevant case feature is valued as accusative (and deleted) as soon as *him* is merged with the transitive verb *helping*, it follows that once we reach the stage of derivation shown in (54) above, the accusative case feature carried by *him* will have been deleted, so making *him* inactive for agreement with T. (We look at accusative case assignment in the next chapter, so will say no more about it for the time being.)

8.7 Agreement and A-movement

So far, we have seen that agreement plays an important role not only in valuing the φ-features of T but also in valuing the case features of nominals. Chomsky (1998, 1999, 2001) goes further and suggests that agreement also plays an important role in A-movement operations. To see why, let's return to consider the derivation of our earlier sentence (5B) *They were arrested.* Assume that the derivation proceeds as sketched earlier, with THEY being merged as the thematic complement of *arrested*, and the resulting VP in turn being merged with the tense auxiliary BE to form the structure (56) below:

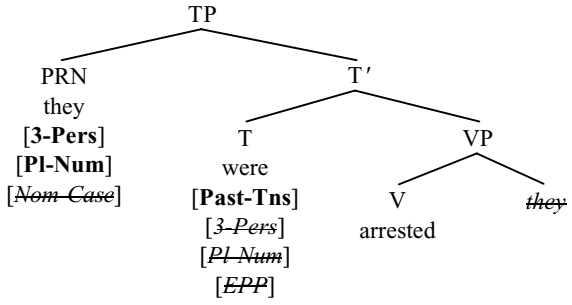
(56)



In (56), [$_T$BE] is an active probe (by virtue of its uninterpretable person and number features) and has an uninterpretable [EPP] feature. It therefore searches for active

nominal goals which can value and delete its person/number features, locating the pronoun THEY (which is active by virtue of its uninterpretable case feature and which has person and number features which match those of BE). Since the matching goal THEY is a definite pronoun, the [EPP] feature of [$_T$ BE] cannot be deleted by merging an expletive in spec-TP, but rather can only be deleted by movement of the goal to spec-TP, in accordance with (45iii): accordingly, THEY moves to become the specifier of BE, thereby deleting the uninterpretable [EPP] feature of BE. Assuming that Feature-Copying, Nominative Case Assignment and Feature-Deletion work as before, the structure which is formed at the end of the TP cycle will be that shown below:

(57)



(To avoid excessive visual clutter, the trace copy of *they* left behind in VP-complement position is shown here simply as ~~*they*~~, but is in fact an identical copy of *they*, containing the same features as *they*. The same typographical convention will be used throughout the rest of this chapter.) The TP in (57) will subsequently be merged with a null declarative-force C, so terminating the syntactic derivation. Since all uninterpretable features have been deleted, the derivation **converges** – i.e. results in a syntactic structure which can subsequently be mapped into well-formed phonetic and semantic representations.

A key assumption underlying the analysis sketched here is that T triggers movement of a nominal goal with which it agrees in person and number. Interesting empirical support for this claim comes from European Portuguese. Costa (2001) notes that in colloquial Portuguese, an intransitive verb used in an unaccusative structure like that below can be either third person singular or third person plural if used with an in-situ postverbal argument as in (58) below:

(58) (a) Fecharam *muitas fábricas*
Closed-3Pl many factories
'Many factories closed'

(b) Fechou *muitas fábricas*
Closed-3Sg many factories
'Many factories closed'

However, the postverbal argument (which originates as the complement of the verb) can only move in front of the verb into spec-TP if the verb (or, more accurately, the associated T-constituent) agrees with the subject in both person and number: cf.

(59) (a) *Muitas fábricas* fecharam
Many factories closed-3Pl
'Many factories closed'

(b) **Muitas fábricas* fechou
Many factories closed-3Sg
'Many factories closed'

This suggests that movement of the italicised nominal from VP-complement position to spec-TP is dependent on full person/number agreement between T and the nominal which it attracts. Costa follows Belletti (1988) in positing that in agreementless sentences like (58b), the postverbal argument is assigned partitive case by the verb (which, in a language in which nouns have a limited case morphology will surface in a form homophonous with the accusative); assignment of case to the complement makes it inactive, and so ineligible to undergo T-agreement – with the result that T surfaces in the agreementless default (third-person-singular) form. It may be that we find a related phenomenon in English sentence pairs such as:

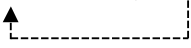
(60) (a) There is *only me* considered suitable
(b) *Only I* am considered suitable

In (60a) the italicised pronoun expression follows the verb *be*, is not assigned nominative case, does not trigger T-agreement and is not raised to spec-TP; by contrast in (60b) the italicised nominal is assigned nominative case, triggers T-agreement, and is moved to spec-TP. Accordingly, sentences like (60) provide empirical support for Chomsky's claim that there is a close association between case, agreement and A-movement.

8.8 EPP in control infinitives

The analysis presented in the previous section assumes that a finite T carries an [EPP] feature which drives A-movement. But what about the kind of infinitival [$_T$ *to*] constituent found in **control** clauses? In the previous chapter, we assumed that infinitival *to* never has an [EPP] feature, and hence that the PRO subject of a control clause like that bracketed in (61a) below remains in situ in spec-VP as in (61b), rather than raising to spec–TP as in (61c):

(61) (a) They don't want [to see you]
(b) They don't want [$_{CP}$ [$_C$ ø] [$_{TP}$ [$_T$ to] [$_{VP}$ PRO [$_V$ see] you]]]
(c) They don t want [$_{CP}$ [$_C$ ø] [$_{TP}$ PRO [$_T$ to] [$_{VP}$ ~~PRO~~ [$_V$ see] you]]]



We noted Baltin's (1995) claim that the *in-situ* analysis (61b) under which PRO remains in situ would account for why *wanna*-contraction is possible in such sentences (yielding *They don't wanna see you*), since there would be no PRO intervening between *want* and *to*. However, Baltin's argument is not entirely convincing. After all, if intervening null constituents block *to* from cliticising onto *want* and if control clauses are CPs, why doesn't the intervening null complementiser in (61b,c) block *wanna*-contraction? If we answer this question (as we did in the previous chapter) by suggesting that *to* first cliticises onto the null complementiser and then the two of them together subsequently cliticise onto *want*, we can handle the relevant data by supposing that only an *overt* subject in

spec-TP (like *who* at the relevant stage of derivation in **Who don't they want to see you?*) in spec-TP blocks cliticisation of *to* onto C, not a null subject like PRO in (61c). What weakens the contraction argument still further is that this kind of contraction is idiosyncratic to the verb WANT (and indeed to the form *want* rather than *wants*, *wanted*, or *wanting*) rather than being associated with all control predicates, and this has led some linguists to suggest that *wanna* should simply be listed in the lexicon as an idiosyncratic form of *want* rather than being the product of a cliticisation operation. (See Boeckx 2000 for an alternative account of *wanna*-contraction.)

In short, the *wanna*-cliticisation argument for saying that PRO remains in situ is potentially flawed. Indeed, there seems to be counter-evidence in support of claiming that PRO *does* in fact move to spec-TP in control infinitives (and hence that control *to* has an EPP feature). Part of the evidence comes from the syntax of constituents like those italicised in (62) below which have the property that they are construed as modifying a bold-printed antecedent which is not immediately adjacent to them in the relevant structure:

(62) (a) **They** were *both* priding themselves on their achievements
(b) **I** don't *myself* think that Svengali was the best choice for England manager
(c) **He** was *personally* held responsible

Both in (62a) is a **floating quantifier** (and *each/all* can be used in a similar fashion); *myself* in (62b) is a **floating emphatic reflexive**; and *personally* in (62c) is an **argument-oriented adverb** (construed as modifying an argument, in this case *he*). In each sentence in (62), the italicised expression is construed as modifying the bold-printed subject of the clause. Contrasts such as those in (63) and (64) below:

(63) (a) **Two republican senators** were *themselves* thought to have been implicated
(b) *There were *themselves* thought to have been implicated **two republican senators**

(64) (a) **Two republican senators** are *both* thought to have been implicated
(b) *There are *both* thought to have been implicated **two republican senators**

suggest that a floating modifier must be c-commanded by its bold-printed antecedent.

In the light of the requirement for a floating modifier to be c-commanded by its antecedent, consider the syntax of the bracketed clauses in the following sentences:

(65) (a) [To *both* be betrayed by their friends] would be disastrous for Romeo and Juliet
(b) [To *themselves* be indicted] would be unfair on the company directors
(c) It was upsetting [to *personally* have been accused of corruption]

In each of these examples, the bracketed clause is a control clause containing a PRO argument. In each case, PRO is the thematic complement of a passive participle (viz. *betrayed/indicted/accused*). Hence, if control *to* has no [EPP] feature and PRO remains in situ, the TP in the bracketed infinitive complement in

(65b) will have the skeletal structure (66a) below, but if control *to* has an [EPP] feature, this will trigger movement of PRO to become the structural subject of *to* – as in (66b):

(66) (a) [$_{CP}$ [$_{C}$ ø] [$_{TP}$ [$_{T}$ to] [$_{AUXP}$ *themselves* [$_{AUX}$ be] [$_{VP}$ [$_{V}$ indicted] **PRO**]]]]
(b) [$_{CP}$ [$_{C}$ ø] [$_{TP}$ **PRO** [$_{T}$ to] [$_{AUXP}$ *themselves* [$_{AUX}$ be] [$_{VP}$ [$_{V}$ indicted] ~~PRO~~]]]]

Given the requirement for a floating emphatic reflexive to be c-commanded by its antecedent, and given that PRO is the intended antecedent of *themselves* in (66), it is clear that (66a) cannot be the right structure, since PRO does not c-command *themselves* in (66a). By contrast, movement of PRO to spec-TP in (66b) means that PRO will indeed c-command *themselves*, so correctly predicting that (66b) is grammatical.

Let's therefore follow Chomsky (1998, 1999, 2001) in positing that control *to* does indeed have an [EPP] feature, triggering raising of PRO to spec-TP. Let's also follow Chomsky in positing that PRO is assigned null case by agreement with a c-commanding T with null (non-finite) tense in much the same way as subjects in tensed clauses are assigned nominative case by agreement with a c-commanding T which has finite (present or past) tense. More specifically, let's assume that *to* in control infinitives contains not only an abstract non-finite tense feature, but also abstract φ-features; and let's further suppose that null case assignment can be characterised informally as follows:

(67) **Null Case Assignment**
An unvalued case feature on a goal is valued as **null** by a probe carrying null (non-finite) tense if probe and goal match in (person and number) φ-features

See Stowell (1982) and Martin (2001) on the tense properties of control *to*, and Martin (2001) for evidence that control *to* has agreement features; but see Bowers (2002) for a different analysis of the case-marking of PRO subjects.

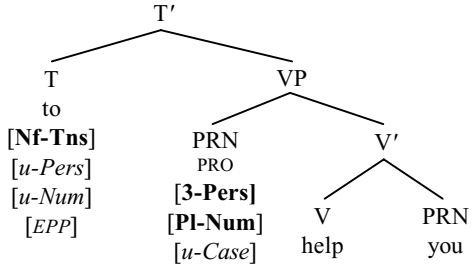
In the light of these assumptions, consider the derivation of the bracketed control clause in:

(68) They have decided [PRO to help you]

Decide is a control predicate (as we see from the fact that (68) is paraphraseable as *They have decided that they will help you*, and from the fact that *decide* does not allow an expletive subject in a sentence like **There has decided to be an enquiry*). Given the VP-Internal Subject Hypothesis, the PRO subject of the bracketed infinitive clause will originate in spec-VP, as the specifier of *help you*. More specifically, the derivation proceeds as follows. The verb *help* merges with its complement *you*, and the resulting V-bar *help you* in turn merges with its PRO subject to form the VP *PRO help you*. Merging control *to* with this VP forms the TP *to help you*. Let's suppose that since PRO refers back to *they* in (68), PRO (as used here) carries the interpretable features [**3-Pers**, **Pl-Num**]; let's also suppose that PRO enters the derivation with an unvalued case feature [*u-Case*]. In addition, let's assume that control *to* carries an interpretable non-finite-tense

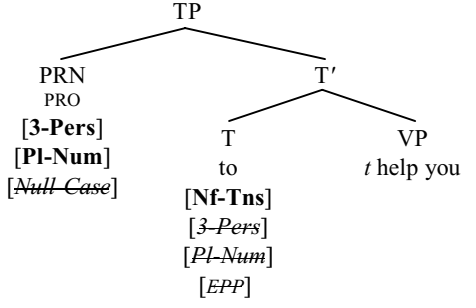
feature [**Nf-Tns**] (denoting an irrealis event which has not yet happened but may happen in the future), and also has uninterpretable (and unvalued) person/number features. Finally (for the reasons given above), let's assume that *to* carries an [EPP] feature in control clauses. Given all these assumptions, merging *to* with [$_{VP}$ *PRO help you*] will form the T-bar (69) below (simplified by showing only features on constituents of immediate concern to us):

(69)



Since *to* is the highest head in the structure and is active (by virtue of its uninterpretable φ-features), it serves as a probe which searches for a goal to value and delete its φ-features. Since *to* c-commands PRO and PRO is active by virtue of its uninterpretable case feature, PRO can serve as a goal for the probe *to*. The unvalued φ-features on the probe are assigned the same third-person-plural values as those on the goal by Feature-Copying (7) and are deleted by Feature-Deletion (14). The unvalued case feature on PRO is assigned the value [*Null-Case*] by Null Case Assignment (67) and deleted by Feature-Deletion (14). Since PRO is a definite pronoun, the [EPP] feature of *to* is deleted by movement of PRO to spec-TP in accordance with the EPP Generalisation (45iii). The result of applying these various operations is to derive the TP (70) below (simplified in a number of ways, e.g. by showing the trace of PRO simply as *t* rather than as a deleted copy of PRO):

(70)



The resulting TP is subsequently merged with the null non-finite complementiser which introduces control clauses. As required, the structure which will serve as input to the semantic component will contain only (bold-printed) interpretable features – all uninterpretable features having been deleted.

A question of incidental detail which arises from the assumption made above that control T assigns null case to a nominal or pronominal expression which it c-commands is why T in (69) cannot assign null case to the pronoun *you* which is the object of the transitive verb *help*, since T c-commands *you* as well as PRO. One answer to this question is that (as we saw in §8.6), transitive verb phrases are **phases**, and hence the Phase Impenetrability Condition (55) allows a T probe to locate a goal on the edge of a transitive VP (like the PRO subject of the VP in (69) above), but not to locate a goal in the c-command domain of a transitive verb (hence not a pronoun like *you* in (69) since this is c-commanded by the transitive phase head *help*). A second answer is that the **Earliness Principle** requires *you* to be assigned case as early as possible in the derivation; and given our assumption in §4.9 that a transitive head assigns accusative case to a noun or pronoun expression which it c-commands, it follows that the case feature carried by *you* will be valued as accusative (and, we suppose, deleted) at the stage of derivation where it is merged as the complement of the transitive verb *help*: and our **Feature Inactivation Hypothesis** (52) tells us that once its case feature is deleted, *you* thereafter becomes inactive for agreement.

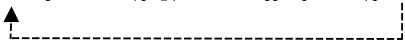
In the previous section, we suggested that a finite T has an [EPP] feature which triggers movement of the closest active matching goal to spec-TP, in conformity with the Attract Closest Principle. In this section, we have suggested that control *to* likewise carries an [EPP] feature triggering movement of the closest active goal to spec-TP. This suggests that we should look to see whether there is some property which finite T and control T share in common which will account for why both of them have an [EPP] feature. One possibility suggested by Chomsky (1999, p. 6) is 'to associate EPP with φ-completeness'. What this would mean is that T has an [EPP] feature only if it has a complete set of φ-features – an idea explored in Nasu (2001, 2002). On the assumption that in a language like English where T probes agree in person and number but not gender with appropriate goals T is φ-complete if it carries person and number features, we can say that a finite T is φ-complete by virtue of carrying person and number features, and the same is true of control *to* under the formulation of Null Case Assignment given in (67) above. However, a natural question to ask in relation to the φ-completeness analysis of [EPP] is whether raising *to* also has an [EPP] feature, and if so whether it is φ-complete or not. This is the question we explore in the next section.

8.9 EPP in other infinitives

In the previous chapter, we posited that raising *to* does not carry an [EPP] feature. This would mean that a sentence such as (71a) below has the skeletal structure (71b), with *he* originating as the thematic subject of *admire* and being raised directly to become the structural subject of *does* (as shown by the dotted arrow):

(71) (a) He does seem to admire her

(b) [$_{CP}$ [$_{C}$ ø] [$_{TP}$ *he* [$_{T}$ does] [$_{VP}$ [$_{V}$ seem] [$_{TP}$ [$_{T}$ to] [$_{VP}$ ~~*he*~~ [$_{V}$ admire] her]]]]]



More specifically, we assumed that *to* in raising structures like (71b) does not have an [EPP] feature, so that *he* does not become the subject of *to* at any stage of derivation. If *to* in raising clauses is assumed to be defective (and hence to lack person and/or number φ-features), this is entirely consistent with Chomsky's suggested generalisation that only a φ-complete T carries an [EPP] feature.

However, Chomsky (2001, fn. 56) argues that (somewhat contrived) sentences like (72) below provide empirical evidence that raising *to* does after all have an [EPP] feature:

(72) John seems to Fred [to appear to himself [to like Mary]]

Here, *himself* refers to *John*, not to *Fred*. This is puzzling if we assume that the antecedent of a reflexive must be an argument locally c-commanding the reflexive (and hence contained within the same TP as the reflexive), since if raising *to* has no [EPP] feature and *John* moves directly from being the subject of the *like* clause to becoming the subject of the *seem* clause, the lefthand bracketed TP containing the reflexive will contain no antecedent for *himself*, and hence we will wrongly predict that sentences like (72) are ill-formed. By contrast, argues Chomsky, if we posit that raising *to* does indeed have an [EPP] feature, *John* will move from being subject of *like Mary* to becoming subject of *to like Mary*, then later becoming subject of *to appear to himself to like Mary*, before finally moving to become the subject of the null T constituent in the *seem* clause. This will mean that a null trace copy of *John* is left behind as the subject of each of the two infinitive clauses, as shown in skeletal form in (73) below:

(73) *John* seems to Fred [~~***John***~~ to appear to himself [~~***John***~~ to like Mary]]

Since the reflexive *himself* is locally c-commanded by the bold-printed trace ~~***John***~~ in (73) within the lefthand bracketed TP containing the reflexive, (73) correctly predicts that *himself* will be interpreted as referring to *John*. (Recall that Chomsky posits that traces are deleted in the phonological component but remain visible in the syntactic and semantic components.) Further evidence that A-movement in raising structures is successive-cyclic is presented in Bošković (2002b).

Sentences like (73) suggest that raising *to* must have an [EPP] feature triggering movement of an argument to spec-TP. But it's important to bear in mind that the [EPP] feature on T works in conjunction with the person/number φ-features of T: more specifically, the [EPP] feature on T triggers movement to spec-TP of an expression which matches one or more of the φ-features of T. It therefore follows that T in raising clauses must carry one or more φ-features if it is to trigger movement of a nominal carrying φ-features of its own. Now it clearly cannot be the case that raising *to* carries both person and number, since if it did we

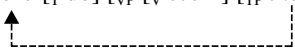
would wrongly predict that raising clauses require a null PRO subject (given that infinitival *to* assigns null case to its subject by (67) when carrying both person and number). The conclusion we reach, therefore, is that raising *to* must carry only *one* φ-feature. But which φ-feature – person or number?

The answer is provided by raising sentences such as the following:

(74) There do seem to remain several problems

On the assumption that raising *to* carries an [EPP] feature requiring it to project a subject, it seems reasonable to posit that expletive *there* will become the specifier of *to remain several problems* at some stage of derivation, and thereafter be raised up (in the manner shown by the arrow in the skeletal structure in (75) below) to become the specifier of *do* on the main-clause TP cycle:

(75) [$_{TP}$ *There* [$_{T}$ do] [$_{VP}$ [$_{V}$ seem] [$_{TP}$ ~~*there*~~ [$_{T}$ to] [$_{VP}$ [$_{V}$ remain] several problems]]]]

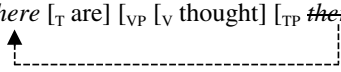


This being so, merging *there* as the specifier of raising *to* on the subordinate clause TP cycle must satisfy the [EPP] feature of *to*. It follows that the φ-feature carried by *to* in (75) must match that carried by expletive *there*. Since we argued in §8.6 that expletive *there* carries person (but not number), it also follows that *to* in (75) must carry a person feature. This being so, the [EPP] feature of raising *to* will require it to project a specifier carrying a person feature, and expletive *there* clearly satisfies this requirement. (Note that the argument goes through irrespective of whether we follow Chomsky 2001 in positing that *there* originates as the specifier of *to*, or Bowers 2002 in assuming that *there* originates as the specifier of *remain* and is subsequently raised up to become the specifier of *to*.)

Our conclusion can be generalised from raising sentences like (74/75) to long-distance passives like (76a) below, involving the movement operation arrowed in (76b):

(76) (a) There are thought to remain several problems

(b) [$_{TP}$ *There* [$_{T}$ are] [$_{VP}$ [$_{V}$ thought] [$_{TP}$ ~~*there*~~ [$_{T}$ to] [$_{VP}$ [$_{V}$ remain] several problems]]]]



Passive *to* (i.e. the kind of *to* found in long-distance passives) cannot carry both person and number features, since otherwise it would wrongly be predicted to require a subject with null case. Since the derivation of (76a) involves a stage at which *there* is the specifier of *to* and since *there* carries person but not number, it seems reasonable to conclude that passive *to* (like raising *to*) likewise carries person but not number.

We can generalise our finding still further to infinitival TPs such as those bracketed in (77) and (78) below:

(77) (a) They were expecting [$_{TP}$ *the visitors* to be met at the airport]
(b) They were expecting [$_{TP}$ *there* to be someone to meet the visitors at the airport]

(78) (a) I will arrange [$_{CP}$ for [$_{TP}$ *the visitors* to be met at the airport]]
(b) I will arrange [$_{CP}$ for [$_{TP}$ *there* to be someone to meet the visitors at the airport]]

The bracketed TPs in (77) are ECM clauses (with the properties noted in §4.8). Since *the visitors* originates as the thematic complement of the passive verb *met* in (77a) but ends up as the subject of [$_{T}$ *to*], it is clear that the head T of the bracketed complement-clause TP must contain an [EPP] feature and at least one φ-feature. Since the infinitive subject can be expletive *there* in (77b), and since *there* carries only person, it follows that the head T of an ECM clause must carry a person feature as well as an [EPP] feature. But if we suppose that a non-finite T which carries a full set of person and number features (like the head T of a control clause) assigns null case to its subject, then it is apparent from the fact that the subject of an ECM clause is an overt constituent and hence does not have null case that the head T of an ECM clause must also be defective, and so carry an [EPP] feature and a person feature, but no number feature. Our conclusion can be generalised in a straightforward fashion to *for*-infinitive structures like those bracketed in (78): if we define ECM structures as structures in which a constituent within TP is assigned case by an external head lying outside the relevant TP, it follows that *for*-infinitives are also ECM structures.

Our argumentation here leads us to the following more general conclusions:

(79) **Feature composition of T in English**
(i) T always carries an [EPP] feature in all types of (finite and non-finite, main and complement) clauses
(ii) T carries a complete set of (person and number) φ-features in finite clauses and non-finite **control** clauses
(iii) T is defective in respect of its φ-features in other types of non-finite clause (e.g. in raising clauses, long-distance passives, and ECM clauses) and carries only person (not number).

And these are essentially the assumptions made in Chomsky (2001).

In the light of the assumptions in (79), consider the derivation of the following sentence:

(80) Several prizes are thought likely [to be awarded]

Since the bracketed infinitive complement in (80) is a defective clause, [$_{T}$ *to*] will carry uninterpretable [EPP] and person features (but no number feature) in accordance with (79i,iii). This means that at the point where *to* is merged with its complement we have the structure shown in skeletal form below:

(81)

[$_{T}$ to]	be awarded [several prizes]
[*u-Pers*]	[**3-Pers**]
[*EPP*]	[**Pl-Num**]
	[*u-Case*]

Since [$_T$ *to*] is the highest head in the structure at this point and is active by virtue of its uninterpretable person feature, [$_T$ *to*] serves as a probe which searches for an active goal and locates *several prizes*, which is active by virtue of its unvalued case feature. The goal *several prizes* values the unvalued person feature of *to* as third person and (by virtue of being φ-complete) deletes it. The unvalued case feature of *several prizes* cannot be valued or deleted by *to*, since *to* is φ-incomplete (by virtue of having no number feature), and only a finite/nonfinite φ-complete T can assign nominative/null case to a goal, and only a φ-complete α can delete a matching feature of ß. The [EPP] feature of *to* is deleted by movement of *several prizes* to spec-TP in accordance with the EPP Generalisation (45iii), thereby deriving the structure (82) below (simplified in various ways, including by showing the deleted trace of *several prizes* without its features):

(82) [several prizes] [$_T$ to] be awarded ~~several prizes~~
[3-Pers] [~~*3-Pers*~~]
[Pl-Num] [~~EPP~~]
[*u-Case*]

Merging the structure (82) with the raising adjective *likely*, merging the resulting AP with the passive verb *thought* and then merging the resulting VP with a finite present-tense T constituent containing BE will derive:

(83) [$_T$ BE] thought likely [several prizes] [$_T$ to] be awarded ~~several prizes~~
[Pres-Tns] **[3-Pers]** [~~*3-Pers*~~]
[*u-Pers*] **[Pl-Num]** [~~EPP~~]
[*u-Num*] [*u-Case*]
[*EPP*]

Because it is the highest head in the structure and is active by virtue of its uninterpretable φ-features, BE serves as a probe which searches for an active goal and locates *several prizes*. By virtue of being φ-complete, the goal *several prizes* values and deletes the uninterpretable person/number features of the probe BE. By virtue of being finite and φ-complete, BE values the unvalued case feature of *several prizes* as nominative, and deletes it. The [EPP] feature of BE is deleted by moving *several prizes* to spec-TP in accordance with (45iii), so deriving:

(84) [several prizes] [$_T$ BE] thought likely ~~several prizes~~ [$_T$ to] be awarded ~~several prizes~~
[3-Pers] **[Pres-Tns]** [~~*3-Pers*~~]
[Pl-Num] [~~*3-Pers*~~] [~~EPP~~]
[~~*Nom-Case*~~] [~~*Pl-Num*~~]
[~~EPP~~]

The resulting TP is subsequently merged with a null declarative complementiser, and BE is ultimately spelled out as *are*. Since all unvalued features have been valued and all uninterpretable features have been deleted, the derivation **converges** (i.e. results in a well-formed structure which can be assigned an appropriate phonetic representation and an appropriate semantic representation).

Now let's return to take another look at the derivation of our earlier sentence (1) *There are thought likely to be awarded several prizes*. Let's adopt Chomsky's TP analysis of expletives and suppose that we have reached the stage of derivation in (81) above, repeated as (85) below:

(85)
[T to]	be awarded	[several prizes]
[*u-Pers*]		[**3-Pers**]
[EPP]		[**Pl-Num**]
		[*u-Case*]

As before, *to* serves as a probe and identifies *several prizes* as an active goal. Since *several prizes* is φ-complete, it can not only value the unvalued person feature of *to* but also delete it, yielding:

(86)
[T to]	be awarded	[several prizes]
[~~*3-Pers*~~]		[**3-Pers**]
[EPP]		[**Pl-Num**]
		[*u-Case*]

Since the goal *several prizes* is an indefinite expression, the [EPP] feature of *to* can be deleted by merging expletive *there* in spec-TP in accordance with the EPP Generalisation (45i), deriving:

(87)
there	[T to]	be awarded	[several prizes]
[*3-Pers*]	[~~*3-Pers*~~]		[**3-Pers**]
	[~~*EPP*~~]		[**Pl-Num**]
			[*u-Case*]

Since *there* is the highest head in the structure and is active by virtue of its uninterpretable person feature, it serves as a probe, and picks out *to* as a matching goal containing a person feature. However, since *to* is defective (in that it has no number feature), it cannot delete the uninterpretable person feature on *there*. (We assume here that *several prizes* cannot serve as a possible goal for *there*, because agreement is a relation between a noun/pronoun expression like *there* and a T constituent like *to*, not a relation between two noun/pronoun expressions like *there* and *several prizes*.)

Merging the TP in (87) with the raising adjective *likely*, merging the resulting AP with the passive verb *thought* and merging the resulting VP with a present-tense T containing BE will derive:

(88)
[T BE]	thought likely	[there]	[T to]	be awarded	[several prizes]
[**Pres-Tns**]		[*3-Pers*]	[~~*3-Pers*~~]		[**3-Pers**]
[*u-Pers*]			[~~*EPP*~~]		[**Pl-Num**]
[*u-Num*]					[*u-Case*]
[EPP]					

At this point, [T BE] is the highest head in the structure and so serves as a probe. Its uninterpretable person and number features make it active, and mean that [T BE] looks for active nominal goals which have person and/or number features.

However, there are two such active nominal goals which are accessible to the probe [$_T$ BE] in (88) – namely the expletive pronoun *there* (active by virtue of its uninterpretable third-person feature) and the quantifier phrase *several prizes* (active by virtue of its uninterpretable case feature, and carrying both person and number features). Both are accessible to [$_T$ BE] in terms of the Phase Impenetrability Condition (55) since neither is c-commanded by a phase head (i.e. by a complementiser or by a transitive verb). Let's suppose (consistent with Chomsky 2001 and with our earlier discussion of (50) above) that when a probe locates more than one active goal, it undergoes simultaneous **multiple agreement** with all active goals accessible to it – in other words, the probe BE simultaneously agrees with both *there* and *several prizes*. The unvalued person feature of BE will be valued as third person via Feature Matching with the third-person goals *there* and *several prizes*; the unvalued number feature of BE will be valued as plural via agreement with the plural goal *several prizes*. The unvalued case feature on the goal *several prizes* will be valued as nominative (and deleted) by the φ-complete probe BE because the two match in person and number and BE carries finite tense. The uninterpretable person/number features of the probe BE can in turn be deleted by the φ-complete goal *several prizes*. In accordance with (45iii) and the Attract Closest Principle, the [EPP] feature of BE attracts the closest active goal (namely *there*) to move to become the specifier of BE (movement resulting in deletion of the [EPP] feature on BE), deriving:

(89)

there	[$_T$ BE]	thought likely	~~there~~	[$_T$ to]	be awarded	[several prizes]
[*~~3-Pers~~*]	**[Pres-Tns]**		[*~~3-Pers~~*]			**[3-Pers]**
	[*~~3-Pers~~*]		[*~~EPP~~*]			**[Pl-Num]**
	[*~~Pl-Num~~*]					[*~~Nom-Case~~*]
	[*~~EPP~~*]					

The resulting structure will then be merged with a null declarative complementiser, and BE will ultimately be spelled out as the third-person-plural present-tense form *are*. As required, all uninterpretable features have been deleted from (89), so only the bold interpretable features are seen by the semantic component.

Note that an important assumption which is incorporated into the analysis presented here is that the φ-features of T agree with every goal which is accessible to them (giving rise to multiple agreement), but that (in consequence of the Attract Closest Principle) the [EPP] feature of T only triggers movement of the **closest** goal to spec-TP.

Under the analysis presented in this section (in which all instances of infinitival *to* carry an [EPP] feature), an important question which arises is how we account for the ungrammaticality of sentences like:

(90) *There are thought likely several prizes to be awarded

Consider first how (90) might be derived, before considering why it is ill-formed. The derivation proceeds along familiar lines until we reach the stage of derivation in (83) above, repeated as (91) below:

(91)

[T BE]	thought likely	[several prizes]	[T to]	be awarded ~~several prizes~~
[**Pres-Tns**]		[**3-Pers**]	[~~*3-Pers*~~]	
[*u-Pers*]		[**Pl-Num**]	[~~*EPP*~~]	
[*u-Num*]		[*u-Case*]		
[*EPP*]				

As before, the case feature of *several prizes* is valued as nominative and deleted by [T BE], and conversely the person/number features of BE are valued and deleted by *several prizes*. Let's suppose that the lexical array contains expletive *there* and that the [EPP] feature of BE is deleted by merging *there* in spec-TP, and that the uninterpretable third-person feature of *there* is deleted by the φ-complete [T BE], so deriving:

(92)

there	[T BE]	thought likely	[several prizes]	[T to]	be awarded ~~several prizes~~
[~~*3-Pers*~~]	[**Pres-Tns**]		[**3-Pers**]	[~~*3-Pers*~~]	
	[~~*3-Pers*~~]		[**Pl-Num**]	[~~*EPP*~~]	
	[~~*Pl-Num*~~]		[~~*u-Case*~~]		
	[~~*EPP*~~]				

(92) is then merged with a null declarative complementiser, and BE is ultimately spelled out as *are*. Since the resulting structure contains no unvalued or uninterpretable features, we expect the corresponding sentence (90) to be well-formed. But it is ungrammatical. Why should this be?

Chomsky's answer is that Merge is a more primitive and less complex operation than Move and that 'Simple operations preempt more complex ones' (Chomsky 1998, p. 18). Merge is a more primitive relation than Move in that other combinatorial systems (like the artificial languages used in mathematics, logic or computer science) employ Merge but not Move. Move is more complex than Merge because it is a composite *agree+copy+merge+pied-pipe* operation. It therefore follows from 'complexity considerations' (Chomsky 1998, p. 18) that spec-TP must be filled by merger if the lexical array (i.e. the set of items taken out of the lexicon to build the relevant sentence structure) contains an expletive, with movement to spec-TP being used only as a last resort (i.e. where the lexical array contains no expletive). As Chomsky (1998, p. 17) puts it, 'Merge preempts the more complex operation Move' (though see Shima 2000 for a dissenting view). Since the sentence in (90) contains expletive *there*, it is clear that the lexical array for (90) includes an expletive. In the light of this observation, let's return to the earlier stage of derivation represented in (81) above, repeated as (93) below:

(93)

[T to]	be awarded	[several prizes]
[*u-Pers*]		[**3-Pers**]
[*EPP*]		[**Pl-Num**]
		[*u-Case*]

Complexity considerations – more explicitly what Chomsky (1999, p. 5) refers to as 'preference of Merge over (more complex) Move' – will require the [EPP] feature of [T *to*] be satisfied by merging expletive *there* in spec-TP, so resulting in (87) above, repeated as (94) below:

(94)

there	[T to]	be awarded	[several prizes]
[*3-Pers*]	[~~*3-Pers*~~]		[**3-Pers**]
	[~~*EPP*~~]		[**Pl-Num**]
			[*u-Case*]

Subsequently, the derivation will proceed through the steps discussed in (88) and (89) above, ultimately deriving the CP structure associated with *There are thought likely to be awarded several prizes*.

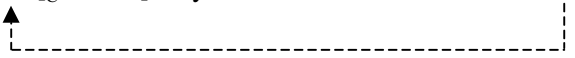
To revert to terminology used in earlier chapters, if T in English always has an [EPP] feature, A-movement will always be a **local** operation which (in complex structures where an argument moves out of one or more lower TP constituents to become the subject of a higher TP) applies in a **successive-cyclic** fashion, with the relevant argument moving to become the subject of a lower TP before going on to become the subject of a higher TP. Since we saw in §5.6 that head movement is also successive-cyclic (in that a moved head moves into the next highest head position within the structure immediately containing it), the greater generalisation would appear to be that all movement is local (and hence successive-cyclic in complex structures), so that any moved constituent moves into the closest appropriate landing site above it (as argued in Rizzi 2001a). If so, we would expect to find that wh-movement is also a local (hence successive-cyclic) operation. And indeed, theoretical considerations suggest that it must be.

In this chapter, we have seen that CPs and transitive VPs are phases, and that the Phase Impenetrability Condition/PIC (55) prevents a constituent which is c-commanded by a complementiser or a transitive verb from being attracted by an external head which c-commands the CP/VP containing the relevant complementiser/transitive verb. PIC turns out to have important consequences for how wh-movement operates in complex sentences such as:

(95) What will they think that he has done?

The wh-pronoun *what* originates as the thematic complement of the transitive verb *done*, and it might at first sight seem as if it moves from being the complement of the transitive verb *done* to becoming the specifier of the C constituent containing the inverted auxiliary *will* in a single step like that shown in highly simplified form in (96) below:

(96) [$_{CP}$ *What* [$_{C}$ *will*+ø] they ~~*will*~~ think that he has done ~~*what*~~]



And indeed, this is precisely what we tacitly assumed in chapter 6. However, a single-step movement operation like that shown in (96) will involve three violations of the Phase Impenetrability Condition, since it involves the bracketed C constituent serving as a probe which attracts the wh-pronoun *what* to move out of a transitive VP headed by *done*, out of a CP headed by *that*, and out of a further transitive VP headed by *think*. The only way of avoiding violation of PIC is for wh-movement to apply in a successive-cyclic fashion, moving *what* first to the

front of the transitive VP headed by *do*, then to the front of the complement clause CP headed by *that*, then to the front of the transitive VP headed by *think*, and finally to the front of the main-clause CP headed by the null complementiser to which the inverted auxiliary *will* adjoins. It would not be appropriate for us to look in more detail at the successive-cyclic nature of wh-movement at this point, until we have taken a closer look at the internal structure of verb phrases in chapter 9 and at the nature of phases in chapter 10: hence we postpone discussion of this until chapter 10.

8.10 Summary

In this chapter, we have taken a look at Chomsky's recent work on case, agreement and A-movement. In §8.2 we saw that agreement plays an integral role in nominative case assignment, in that nominative case is assigned to a nominal which agrees in person and number with a finite T. In §8.3 we argued that some features enter the derivation already valued (e.g. the tense feature of T and the person/number φ-features of nominals), whereas others (e.g. the φ-features of T and the case feature of nominals) are initially unvalued and are assigned values in the course of the derivation by operations like Feature-Copying (7) and Nominative Case Assignment (9). In §8.4, we claimed that interpretable features enter the derivation already valued, whereas those features which are initially unvalued are uninterpretable. We saw that agreement and case-marking involve a relation between an active probe and an active goal, and that probe and goal are only active if they carry one or more uninterpretable features (e.g. uninterpretable φ-features or case features). We also saw that uninterpretable features have to be deleted in the course of the derivation by a Feature-Deletion operation (14), in order to ensure that they do not feed into the semantic component and thereby cause the derivation to crash (because they are **illegible** in the semantic component), and that only a φ-complete α can delete an uninterpretable feature of ß. In §8.5, we suggested that expletive *it* enters the derivation with uninterpretable third-person and singular-number features, and that these value, delete and in turn are deleted by those of the auxiliary in sentences such as *It is said that he has taken bribes*. However, we noted that weather *it* is quasi-referential, and may originate as an argument internally within VP. In §8.6, we looked at Chomsky's claim that *there* is merged directly in spec-TP, and serves as a probe whose uninterpretable third-person feature is deleted via agreement with a φ-complete T. We noted that such an analysis has consequences (for the nature of probes and the inactivation of features) which may not seem desirable, and outlined an alternative analysis of expletives as originating within VP. In §8.7 we outlined Chomsky's agreement-based theory of movement under which movement involves an agreement relation between an active probe with an [EPP] feature and an active goal, and we suggested in (45) that the [EPP] feature of T can be satisfied either by merger of an expletive in spec-TP, or by movement of the closest active matching

goal to spec-TP, with merger/movement of the relevant constituent in spec-TP deleting the [EPP] feature of T. In §8.8 we looked at the syntax of control infinitives, claiming that their PRO subject is assigned null case via agreement with a φ-complete T carrying null (non-finite) tense. We went on to argue that data relating to the distribution of floating modifiers suggest that T in control clauses has an [EPP] feature which triggers movement of PRO to spec-TP. In §8.9 we argued that T in other kinds of infinitive clause (e.g. the infinitival complements of raising, passive and ECM predicates) is defective in that although it carries uninterpretable [EPP] and person features (the latter serving to make T **active**), it lacks the number feature carried by a φ-complete T in finite and control clauses. We saw that such an analysis entails that A-movement takes place in a successive-cyclic fashion, with the moved argument being raised to become the subject of a lower TP before raising to become the subject of a higher TP. We went on to consider the possibility that all movement operations are local (and hence apply in a successive-cyclic fashion in complex structures) and noted that this implies that A-bar movement operations like wh-movement are also successive-cyclic in complex sentences (but said we would postpone detailed discussion of this until chapter 10).

Workbook section

Exercise 8.1

Discuss the derivation of the following sentences.

1 There remains little hope of finding any survivors
2 There are expected to remain some problems
3 Several mineworkers are thought to have died
4 They are considered likely to want to enter the race
5 It would be unthinkable to compromise yourself
6 He appears to be thought to be certain to win the race
7 He is hoping to be promoted
8 It is rumoured that there have been several riots
9 It is thought to be likely that some houses have collapsed
10 There seem certain to remain some problems

In addition, say why the derivation of each of the following sentences crashes (informal paraphrases of the intended meaning being given where this may be unclear):

11 *There are expected some problems to remain (= 'There are expected to remain some problems')
12 *Some problems are expected there to remain (= 'There are expected to remain some problems')
13 *It is expected there to remain some problems (= 'There are expected to remain some problems')
14 *There are rumoured that several prisoners were recaptured
15 *They appear were detained (= 'It appears that they were detained')
16 *He is believed has quit his job (= 'It is believed that he has quit his job')

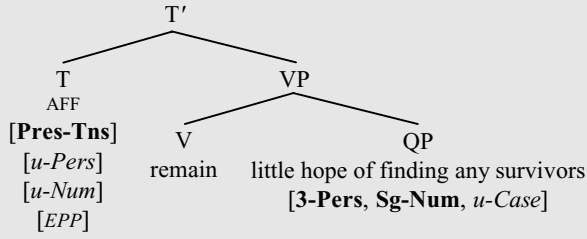
Helpful hints

Assume (as in §§4.6–4.7) that all finite clauses and all control infinitive clauses (i.e. infinitive clauses which are the complement of a CONTROL predicate like *decide*) are canonical clauses and hence CPs containing a T headed by a φ-complete T with person, number and [EPP] features. By contrast, assume (as in §4.8) that seemingly subjectless infinitive clauses which are the complement of a raising predicate like *seem/likely* or a passive participle like *expected* are defective clauses, and hence are TPs headed by a defective T which has person and [EPP] features, but no number feature. Be careful not to confuse raising structures like *It's likely to rain* with control structures like *It's easy to make a mistake*: note that in the first structure we can have an expletive *there* subject (*There's likely to be a strike*), but not in the second (**There's easy to be a mistake*). Assume that infinitival *be* is the head AUX constituent of an AUXP when followed by a passive participle, but the head V of a VP when followed by an adjective. Finally, assume that the preference for Merge over Move holds only internally within a given CP (i.e. where an expletive and a nominal which are competing for the same spec-TP position are both contained in the same CP). Do any of these sentences prove problematic for the analysis in the text – and if so, why?

Model answer for sentence 1

If we assume Chomsky's TP analysis of expletives, sentence 1 will be derived as follows. The noun *hope* is merged with its PP complement *of finding any survivors* (whose structure need not concern us here) to form the NP *hope of finding any survivors*. This NP is merged with the quantifier *little* to form the QP *little hope of finding any survivors*. This QP is merged as the complement of the unaccusative verb *remain* to form the VP *remain little hope of finding any survivors*. This in turn is merged with an affixal finite T constituent (below denoted as AFF) to form the structure shown in simplified form in (i) below:

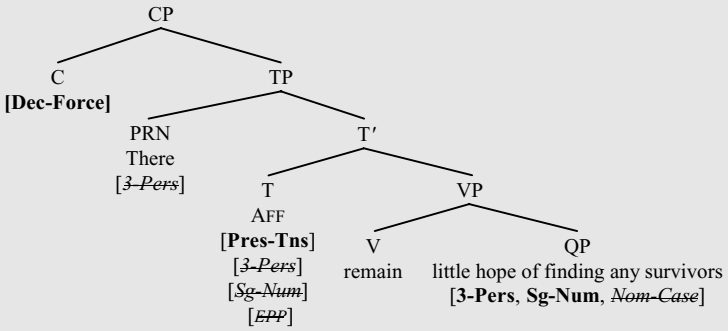
(i)



The affixal T serves as a probe because it is the highest head in the structure, and because its uninterpretable person/number features make it active. In accordance with the Earliness Principle, T immediately searches for an active goal, locating the QP *little hope of finding any survivors* (which is active by virtue of having an uninterpretable case feature). The T probe values the case feature on the QP goal as nominative via the Nominative Case Assignment operation (9) in the main text, and deletes it via Feature-Deletion (14). Conversely, the QP goal values the unvalued φ-features on the T probe as third person singular via Feature-Copying (7) and deletes them via Feature-Deletion (14). If the lexical array for the relevant CP (i.e. the set of constituents out of which the CP in question is formed) includes expletive *there*, preference of Merge over Move will mean that the [EPP] requirement for T to have a specifier must be satisfied by

merging *there* in spec-TP, thereby deleting the [EPP] feature of T in accordance with (45i); the uninterpretable third-person feature of *there* will be deleted by the φ-complete affix in T. Merging the resulting TP with a null complementiser marking the sentence as declarative in force derives:

(ii)



All the features in (ii) feed into the PF component, and since there are no unvalued features in (ii), the relevant structure can be assigned an appropriate PF representation: since there is no overt auxiliary in T on which the Tense Affix (AFF) containing the person/number/tense features of T can be spelled out, the relevant affix is lowered onto the verb *remain* (by the morphological operation of Affix Hopping) in the PF component, so that the verb ultimately surfaces in the third-person-singular present-tense form *remains*. Since all features in (ii) are valued, (ii) can be mapped into an appropriate PF representation; and since all (italicised) uninterpretable features have been deleted, (ii) can also be mapped into an appropriate semantic representation.

Exercise 8.2

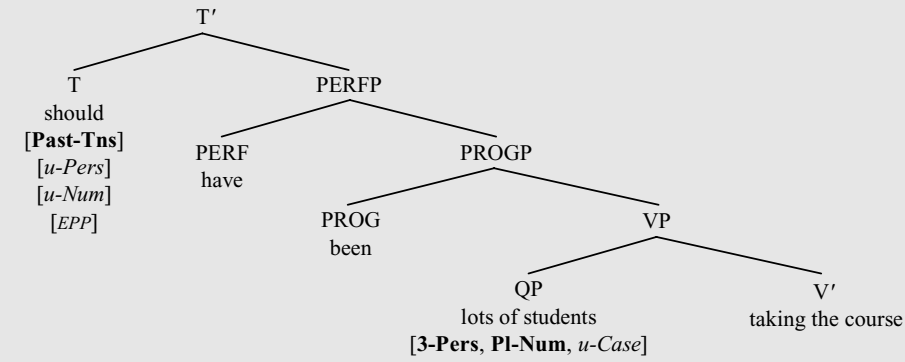
Discuss the derivation of following Belfast English sentences (kindly supplied to me by Alison Henry):

1 There should have been lots of students taking the course
2 There should have lots of students been taking the course
3 There should lots of students have been taking the course
4 There have seemed to be lots of students enjoying the course
5 There have seemed lots of students to be enjoying the course
6 There have lots of students seemed to be enjoying the course

Model answer for sentence 1

If we assume Chomsky's TP analysis of expletive *there*, sentence 1 will have the following (simplified) derivation. The transitive verb *taking* merges with its DP complement *the course* to form the V-bar *taking the course*. This is in turn merged with its subject QP *lots of students* (whose internal structure need not concern us here) to form the VP *lots of students taking the course*. This then merges with the progressive auxiliary *been* to form the progressive auxiliary projection (PROGP) *been lots of students taking the course*. This in turn is merged with the perfect auxiliary *have* to form the Perfect Auxiliary Projection (PERFP) *have been lots of students taking the course*. The resulting PERFP is in turn merged with a finite T constituent containing the past-tense modal auxiliary *should*, so deriving the T-bar shown in simplified form in (i) below (where only the features of those constituents of immediate interest to us are shown):

(i)



By virtue of its uninterpretable person and number features, [$_T$ *should*] serves as a probe and identifies the QP *lots of students* as the only accessible active goal. (Since a transitive VP is a phase, the Phase Impenetrability Condition prevents T from accessing any constituent of a transitive VP other than its specifier and head – and hence prevents the VP complement *the course* from being a goal for T.) Accordingly, *lots of students* values (as third person plural) and deletes the person/number features of *should*, and conversely *should* values (as nominative) and deletes the case feature of *lots of students*. Since the goal *lots of students* is an indefinite expression, the [EPP] feature carried by [$_T$ *should*] can be deleted by merging expletive *there* in spec-TP in accordance with the EPP Generalisation (45i) in the main text – and indeed, preference of Merge over Move would dictate that we need to use an expletive subject if (as in 1) we have an expletive pronoun in our lexical array. Merging expletive *there* in spec-TP will delete the [EPP] feature of T, and conversely the uninterpretable person feature carried by *there* can be deleted by the φ-complete T constituent *should*, so deriving the simplified structure:

(ii)

There	*should*	have been	*lots of students*	taking the course
[~~*3-Pers*~~]	**[Past-Tns]**		**[3-Pers]**	
	[~~*3-Pers*~~]		**[Pl-Num]**	
	[~~*Pl-Num*~~]		[~~*Nom-Case*~~]	
	[~~*EPP*~~]			

The resulting TP will subsequently be merged with a null declarative C. Since all unvalued features have been valued and all uninterpretable features deleted, the resulting derivation is convergent (in that it can be mapped into appropriate phonetic and semantic representations).

Helpful hints on sentences 2–6

Discuss the problems posed for Chomsky's Prefer-Merge-Over-Move principle by some of the sentences 2–6, and also the problems posed for the assumption made throughout our text so far that only C (in wh-clauses) and T (in all types of clause) have an [EPP] feature. Consider the possibility of an alternative account under which languages (and language varieties) may differ in respect of an EPP parameter in relation to what kind of heads carry an [EPP] feature.